This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

Compressive Sensing Image Restoration using Adaptive Curvelet Thresholding and Nonlocal Sparse Regularization

Nasser Eslahi, *Student Member, IEEE,* Ali Aghagolzadeh, *Senior Member, IEEE*

***Abstract*—Compressive sensing (CS) is a recently emerging technique and an extensively studied problem in signal and image processing, which suggests a new framework for simultaneous sampling and compression of sparse or compressible signals at a rate significantly below the Nyquist rate. Maybe, designing an effective regularization term reflecting the image sparse prior information plays a critical role in CS image restoration. Recently, local smoothness and nonlocal self-similarity have both led to superior sparsity prior for CS image restoration. In this paper, first, an adaptive curvelet thresholding criterion is developed, trying to adaptively remove the perturbations appeared in recovered images during CS recovery process, imposing sparsity. Furthermore, a new sparsity measure called joint adaptive sparsity regularization (JASR) is established, which enforces both local sparsity and nonlocal 3D sparsity in transform domain, simultaneously. Then, a novel technique for high-fidelity CS image recovery via JASR is proposed—CS-JASR. To efficiently solve the proposed corresponding optimization problem, we employ the split Bregman iterations. Extensive experimental results are reported to attest the adequacy and effectiveness of the proposed method comparing with the current state-of-the-art methods in CS image restoration.**

***Index Terms*—Compressive sensing, sparse recovery, adaptive curvelet thresholding, nonlocal self-similarity.**

I. Introduction

THE emerging field of compressive sensing (CS) [1], [2]– also called compressed sensing or compressive sampling– integrates joint signal acquisition and compression into a unified approach, providing an alternative to the well-known Shannon/Nyquist sampling theory when the underlying signal is known to be sparse or compressible. It also permits that under certain conditions, the original signal can be reconstructed properly from a small set of measurements using sparsity-promoting nonlinear recovery algorithms.

Suppose that $\boldsymbol{u} \in \mathbb{R}^n$ is a finite length signal. It is said to be *sparse* (or *compressible*) if $\boldsymbol{u}$ can be written as a superposition of a small number of vectors taken from a known sparsifying transform domain basis $(t = n)$ or frame $(t > n)$ $\boldsymbol{\Psi}_s \in \mathbb{R}^{t\times n}$, such that $\boldsymbol{\Psi}_s\boldsymbol{u}$ contains only a small set of nonzero (or significant) entries. According to the CS theory, rather than observing directly the original signal $\boldsymbol{u}$, a number of $m \ll n$ linear measurements are acquired through the following random linear projection:

$$\boldsymbol{f} = \boldsymbol{\Phi}\boldsymbol{u} + e, \tag{1}$$

where $\boldsymbol{f} \in \mathbb{R}^m$ is the observation/measurement vector, $\boldsymbol{\Phi} \in \mathbb{R}^{m\times n}$ is a sensing/measurement matrix and $\boldsymbol{e} \in \mathbb{R}^{m\times 1}$ denotes additive noise (or measurement error). Since the number of unknowns is much more than the observations, clearly we are not able to recover every $\boldsymbol{u}$ from $\boldsymbol{f}$ and it is generally considered as an *ill-posed* problem. However, if $\boldsymbol{u}$ is sufficiently sparse (or compressible) in a sparsifying transform domain $\boldsymbol{\Psi}_s$, and let the sensing matrix $\boldsymbol{\Phi}$ be highly incoherent with the sparsifying matrix $\boldsymbol{\Psi}_s$, then the exact recovery of $\boldsymbol{u}$ is possible [1]. In order to obtain an estimate with a reasonable accuracy and robustness to noise, one must often rely on prior information about the original signal. The estimation of $\boldsymbol{u}$ is then formulated as a regularization-based optimization problem which incorporates this prior information, i.e.,

$$\min_{\boldsymbol{u}}\{\frac{1}{2}\|\boldsymbol{f} - \boldsymbol{\Phi}\boldsymbol{u}\|_{\ell_2}^2 + \lambda\mathcal{R}(\boldsymbol{u})\}. \tag{2}$$

The term $\mathcal{R}(\boldsymbol{u})$ could be various choices, e.g., $\|\boldsymbol{u}\|_{\ell_p}$, $\|\boldsymbol{\Psi}_s\boldsymbol{u}\|_{\ell_p}$ where $\ell_p \in \{\ell_0, \ell_1\}$, total variation $TV(\boldsymbol{u})$ [3] or Bregman distance [4]. The first term in (2) is a penalty that represents the closeness of the solution to the observed scene and quantifies the prediction error with respect to the measurements. The second term in (2) is a regularization term that represents *a priori* information about the original scene (e.g., sparsity) and also it is designed to penalize an estimate that would exhibit the expected properties. Also, λ is a regularization parameter that balances the contribution of both terms. This minimizing problem can be solved easily by various methods, such as iterative shrinkage/thresholding (IST) methods (e.g., [5]-[7]) and Bregman iterative algorithms (e.g., [8], [9]).

Since exploiting a prior knowledge of the original signal plays a substantial role in image inverse problems and CS theory, numerous efforts have been made to develop an effective regularization term $\mathcal{R}(\boldsymbol{u})$ and more realistic models, to reflect the image prior knowledge. Sparse regularization is a popular class of priors to model natural signals and images. Most of the sparsity-based approaches that have been introduced so far in the literature can be assigned to two main categories, namely the *analysis*-based and *synthesis*-based sparse regularization [10]. Analysis-based priors suggest that a signal is sparse in an analysis domain, while

N. Eslahi is with the Department of Signal Processing, Tampere University of Technology, Tampere 33720, Finland (e-mail: nasser.eslahi@tut.fi)

A. Aghagolzadeh is with the Department of Electrical and Computer Engineering, Babol University of Technology, Babol 47148-71167, Iran (e-mail: aghagol@nit.ac.ir)

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

synthesis-based priors seek a representation of the signal as a linear combination of small number of columns/atoms from a synthesis dictionary. Conventional CS recovery approaches exploit the sparsity of a signal by a sparsity-promoting regularization term like $\|\mathbf{\Psi}_s \boldsymbol{u}\|_{\ell_p}$, which incorporating it into (2) forms an analysis formulation that is common in many inverse problems. Also, natural signals and images are well-known to be approximately sparse in analytical transform domains, e.g., discrete cosine transform, wavelets, ridgelets [11], curvelets [12], and contourlets [13]. Compared with the isotropic elements of wavelets, the needle-shaped elements of curvelets [12], [14] possess highly directional sensitivity and anisotropy which lead to an optimal sparse representation of object with discontinuities along smooth curves; thereby gaining good performance in CS image restoration [15]-[17].

The classic smoothing regularization terms, such as the quadratic Tikhonov regularization [18] and the total variation (TV) [3] regularization utilize local structural patterns and are built based on the assumption that images are locally smooth except at edges. More specifically, despite the fact that the later model leads to a reconstruction that characterizes sharp and well-preserved image edges, they both favor piecewise-constant image structures, and hence tend to smooth much the image details and fine structures (resulting in staircase artifacts and contrast losses), since they only exploit the local structural regularity property, neglecting the nonlocal statistics of images [19], [20].

In addition to the sparsity and the local statistics, the nonlocal self-similarity is another significant property emerging in natural images, which depicts the repetitiveness of the higher level patterns (e.g., textures and structures) globally positioned in images. That is to say, images are often comprised of localized patterns that repeat themselves at distant locations in the image domain. That being so, nonlocal regularizers can effectively model long-range dependencies and yield improvements in reconstruction results [21]. Inspired by the success of nonlocal means (NLM) filtering for image denoising [22], several nonlocal regularization-based methods have also been proposed for various image processing applications [19]-[21], [23]-[39], and also CS image restoration [40]-[49].

A. Related Work on CS via Nonlocal Sparse Regularization

In recent works, the sparsity and the nonlocal self-similarity properties are usually combined into the final cost functional of image restoration solution to yield better performance. A combinational regularization parameter, using a reweighted TV and a weighted-based nonlocal sparse constraint, for CS image recovery is proposed in [41]. The work in [43] proposed an adaptive sparsity regularization term for CS image recovery process, which incorporated the local piecewise autoregressive model and a weighted-based nonlocal self-similarity constraint. In [44], the sparsity regularization parameters (which are locally estimated) together with a weighted-based nonlocal self-similarity constraint are incorporated into the overall cost functional of image restoration solution to improve the image quality. A model-assisted adaptive recovery of CS (MARX-PC) is proposed in [45], which exploits both the local structural sparsity and nonlocal self-similarity, leading to an efficient CS recovery scheme. A nonlocal CS image recovery method (NLCS) is proposed in [46], which exploits the nonlocal patch correlation and the local piecewise smoothness prior in natural images, to reduce the sampling rate of CS images. In [47], a nonlocal low-rank regularization approach, toward exploiting the structured sparsity for CS image recovery is proposed. The proposed model in [47] consists of two components: patch grouping for characterizing the self-similarity of the signal and low-rank approximation for sparsity enforcement. The work in [48] proposed a strategy for CS image recovery via collaborative sparsity (RCoS) modeling. The local 2D sparsity and the nonlocal 3D sparsity are simultaneously imposed in RCoS, enabling a natural image to be highly sparse in an adaptive space-transform domain.

B. Contributions

In [17], we proposed a block-based CS image recovery approach based on iterative curvelet thresholding (ICT), namely BCS-ICT, which considers (2) with regularization term $\mathcal{R}(\boldsymbol{u}) = \|\mathbf{\Psi}\boldsymbol{u}\|_{\ell_1}$ and disceret curvelet transform (DCuT) [50] as the sparsifying transform. BCS-ICT employs an iterative shrinkage approach which is highly dependent on the empirical-based linear decay thresholding criterion that is non-adaptive and sensitive to noise. To combat this drawback, we propose an *adaptive curvelet thresholding* (ACT) criterion which tries to adaptively remove the perturbations appeared in recovered images during the CS recovering process, by removing the insignificant curvelet coefficients, and imposing sparsity. To delineate, in the proposed statistical approach, ACT can be considered as a denoising/shrinkage procedure in curvelet domain via estimation of clean curvelet coefficients from available rough estimated data, in each iteration, with Bayesian estimation technique. Accordingly, we extend our previous work [17], BCS-ICT, by employing ACT within its framework to promote its efficiencies and lead to a simple and effective CS image recovery method.

Inspired by the promising results of the previously-mentioned techniques in CS image recovery via nonlocal sparse regularizations, in this paper, the nonlocal self-similarity and the local sparsity prior are both incorporated in a combinational regularization term, to adopt in a regularization-based framework for CS image restoration. To be more precise, we introduce a new sparsity measure, called *joint adaptive sparsity regularization* (JASR). The main hypothesis underlying the proposed JASR is that it enforces both the local sparsity constraint and the nonlocal 3D sparsity in transform domain, in a unified statistical approach, concurrently, suggesting a powerful mechanism for characterizing the structured sparsities of natural images. In fact, the local sparsity depicts the local smoothness redundancies exploited by DCuT [50], which is boiled down to ACT, and the nonlocal 3D sparsity corresponds to the nonlocal self-similarity constraint achieved by a nonlocal statistical sparse modeling closely related to the ones proposed in [20] and [48]. Unlike those introduced in [20] and [48], our proposed *nonlocal statistical sparse modeling* (NSSM) has some functional modifications which makes it

TABLE I: THE IMPORTANT NOTATIONS USED IN THIS PAPER

Notation:	*Description*:
$\boldsymbol{u}$	original signal vector (desired signal)
$\boldsymbol{\Phi}$	measurement/sensing matrix
$\boldsymbol{f}$	measurement/observation vector
$\boldsymbol{\Psi}_s$	sparse transform domain basis or frame
$\boldsymbol{\Psi}$	forward curvelet transform matrix
$\boldsymbol{\vartheta}$	sparse transform coefficient vector
$j, l, \mathbf{k}$	the curvelet triple index of scale, orientation, and spatial location
$\Psi_{\mathrm{N3D}}(\cdot)$	the nonlocal 3D sparsity (N3D) operator
$\Psi_{\mathrm{NLSM}}(\cdot)$	the nonlocal statistical modeling (NLSM) operator
$\Psi_{\mathrm{NSSM}}(\cdot)$	the nonlocal statistical sparse modeling (NSSM) operator
$\Omega_{\mathrm{N3D}}(\cdot)$	the inverse operator of $\Psi_{\mathrm{N3D}}(\cdot)$
$\Omega_{\mathrm{NSSM}}(\cdot)$	the inverse operator of $\Psi_{\mathrm{NSSM}}(\cdot)$
$\boldsymbol{u}_{p_{l'}}$	the overlapped patch extracted from $\boldsymbol{u}$ at location l'
$\boldsymbol{G}_{\boldsymbol{u}_{p_{l'}}}$	the 3D array of stacked matched patches (group)
$\mathcal{T}^{3D}(\cdot)$	the operator of the orthogonal 3D transform
$\boldsymbol{\Theta}_{\boldsymbol{u}}$	the column vector of all the transform coefficients of image $\boldsymbol{u}$
k	iteration number
Ψ_{DCuT}	image local smoothness prior via DCuT
$\hat{\boldsymbol{u}}, \hat{\hat{\boldsymbol{u}}}, \tilde{\boldsymbol{u}}, \breve{\boldsymbol{u}}$	estimations for $\boldsymbol{u}$
$\boldsymbol{I}_n$	identity matrix of size $n \times n$

much more superior for CS image recovery—as introduced and examined in Sections IV and VI-C, respectively.

We further propose a novel technique for high-fidelity *CS image restoration using JASR* (we refer to it as CS-JASR). The proposed CS-JASR is formulated in a form of minimization functional under regularization-based framework. Based on split Bregman framework [8], [9], a powerful method for solving various variational models, an efficient alternating minimization algorithm is developed to attack the above severely underdetermined inverse problem efficiently.

Extensive numerical results on benchmark test images clearly demonstrate that our proposed methods considerably outperform many conventional and state-of-the-art techniques for CS image restoration, and also exhibit good convergence property. To evaluate our simulation results, we use two applicable quality assessors, the peak signal-to-noise ratio (PSNR) in dB and the structural similarity (SSIM) [51].

C. Notations

To make our mathematical formulations clear, we reserve a bold uppercase to denote a matrix, a bold lowercase to denote a column vector, and regular lowercase or uppercase to denote a scalar. Besides, Table I summarizes some important notations used in the following sections.

D. Outline

The rest of this paper is organized as follows. Two recently relevant proposed models for nonlocal self-similarity are briefly introduced in Section II. Also, a brief background on curvelet transform accompanied by our previous work on CS image recovery via curvelet transform are provided there. The proposed ACT along with the extension of our previous work via ACT are introduced in Section III. The proposed regularization term, JASR, and its relation to the previous works are described and discussed in Section IV. Section V shows how JASR is incorporated into the framework of image CS restoration, giving the implementation details of solving the ensuing optimization problem. Numerical results and comparisons for our proposed methods are given in Section VI and finally, Section VII concludes the paper.

II. BACKGROUNDS AND PRELIMINARIES

A. Nonlocal 3D Sparsity and Nonlocal Self-Similarity Modeling in Transform Domain

Motivated by the success of NLM [22] in self-similarity and BM3D [52] in image restoration, Zhang *et al.* [48] proposed a model for nonlocal self-similarity prior information, which is employed efficiently in CS image restoration. The proposed model explores the nonlocal self-similarity by means of the sparsity of the transform coefficients, which are obtained by transforming the 3D array generated by stacking similar image patches. To elucidate on, first, the image $\boldsymbol{u}$ (of size $\sqrt{n} \times \sqrt{n}$) is divided into P overlapped patches of equal sizes $\boldsymbol{u}_{p_{l'}} \in \mathbb{R}^{B_p}$ at location $l', l' = 1, 2, \cdots, P$. Then, the set including the c best matched patches (based on Euclidean distance) in a search window of size $\omega \times \omega$ is defined as $\boldsymbol{S}_{\boldsymbol{u}_{p_{l'}}} = \{\boldsymbol{S}_{\boldsymbol{u}_{p_{l'}} \otimes q}\}_{q=1}^{c}$. For each $\boldsymbol{S}_{\boldsymbol{u}_{p_{l'}}}$, the c best matched patches belonging to $\boldsymbol{S}_{\boldsymbol{u}_{p_{l'}}}$ are stacked into a 3D array to form a group $\boldsymbol{G}_{\boldsymbol{u}_{p_{l'}}}$. Next, the transform coefficients of each group is found by applying a 3D transform to the group, i.e., $\mathcal{T}^{3D}(\boldsymbol{G}_{\boldsymbol{u}_{p_{l'}}})$, followed by arranging them in a lexicographic order to form $\boldsymbol{\Theta}_{\boldsymbol{u}} \in \mathbb{R}^{K_\Theta}$ (the column vector of all the transform coefficients of image $\boldsymbol{u}$), where $K_\Theta = B_p * c * P$. At last, the number of nonzero coefficients is used to measure the N3D sparsity of this patch while the N3D sparsity of the whole image is obtained by summing all the ones of each patch [48]. Thus, the N3D for self-similarity in transform domain can be mathematically expressed as

$$\Psi_{\mathrm{N3D}}(\boldsymbol{u}) = \|\boldsymbol{\Theta}_{\boldsymbol{u}}\|_{\ell_0} = \sum_{l'=1}^{P} \|\mathcal{T}^{3D}(\boldsymbol{G}_{\boldsymbol{u}_{p_{l'}}})\|_{\ell_0}. \tag{3}$$

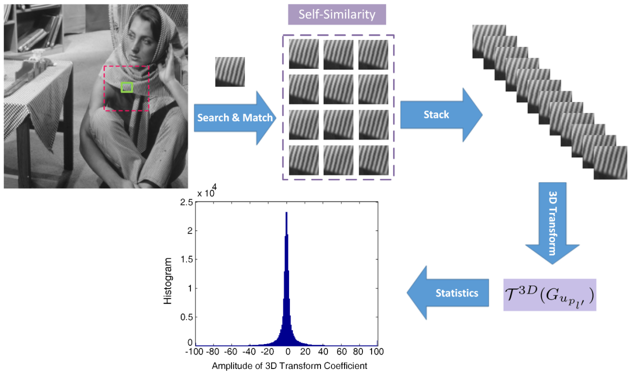
After obtaining $\boldsymbol{\Theta}_{\boldsymbol{u}}$, it is split into P groups (3D arrays) of transform coefficients, which are then inverted to generate estimates for each patch in the group. The patch-wise estimates are returned to their original positions and the final estimated image is achieved by averaging all of the above patch-wise estimates [48]. To put it simply and briefly, after obtaining $\boldsymbol{\Theta}_{\boldsymbol{u}}$, the new estimate of $\boldsymbol{u}$ is achieved by $\hat{\boldsymbol{u}} = \Omega_{\mathrm{N3D}}(\boldsymbol{\Theta}_{\boldsymbol{u}})$.

Besides, inspired by the success of NLM and BM3D, a nonlocal statistical modeling (NLSM) for self-similarity in 3D transform is proposed by Zhang *et al.* [20], which is used in image restoration application—except CS. Fig. 1 gives an illustrative overview of the NLSM for self-similarity in 3D transform domain. As can be seen, the NLSM, which is closely related to N3D, characterizes the nonlocal self-similarity of natural images by means of the distribution of the transform coefficients $\boldsymbol{\Theta}_{\boldsymbol{u}}$, i.e., $\Psi_{\mathrm{NLSM}}(\boldsymbol{u}) = \|\boldsymbol{\Theta}_{\boldsymbol{u}}\|_{\ell_1}$.

B. Curvelet Transform

Traditional wavelets, irrespective of their good performance in representing objects with point singularities, fail to repre-

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing



(a)

Fig. 1: Illustration of NLSM for self-similarity in 3D transform domain [20].

sent line or curve singularities since they ignore geometric properties of structures and do not exploit the regularity of edges. Thus, wavelet-based compression, denoising or structure extraction become computationally inefficient for geometric features with line and surface singularities [15].

The curvelet transform [12], [14] is a multi-scale directional transform which provides a new multi-resolution representation with several features that are superior to existing representation, i.e., wavelets and steerable pyramids. It represents edges and singularities along curves more precisely with the anisotropic needle-shaped elements which possess super directional sensitivity and smooth contours capturing efficiently. Such elements are very effective in representing curve-like edges and texture-like features [15]. Unlike wavelets, curvelets are described by a triple index: 1) scale $2^{-j}, j \in \mathbb{N}_0$; 2) equi-spaced sequence of rotation angle $\theta_{j,l} = 2\pi l.2^{-\lfloor \frac{j}{2} \rfloor}$, $l = 0, 1, \cdots, 2^{\lfloor \frac{j}{2} \rfloor} - 1$ such that $0 \leq \theta_{j,l} \leq 2\pi$; and 3) position $\mathbf{x}_{\mathbf{k}}^{(j,l)} = \mathbf{R}_{\theta_{j,l}}^{-1}(k_1 2^{-j}, k_2 2^{-\lfloor \frac{j}{2} \rfloor})^T; \mathbf{k} = (k_1, k_2) \in \mathbb{Z}^2$, where $\mathbf{k}$ is the sequence of translation parameters and $\mathbf{R}_{\theta_{j,l}}$ denotes the rotation matrix with angle $\theta_{j,l}$, defined as

$$\mathbf{R}_{\theta_{j,l}} = \begin{pmatrix} \cos\theta_{j,l} & \sin\theta_{j,l} \\ -\sin\theta_{j,l} & \cos\theta_{j,l} \end{pmatrix}. \tag{4}$$

The curvelet elements (family of curvelets) are defined by

$$\Psi_{j,l,\mathbf{k}}(\mathbf{x}) := \Psi_j(\mathbf{R}_{\theta_{j,l}}(\mathbf{x} - \mathbf{x}_{\mathbf{k}}^{(j,l)})), \mathbf{x} = (\mathrm{x}_1, \mathrm{x}_2) \in \mathbb{R}^2 \tag{5}$$

where Ψ_j are smooth functions with compact support on wedges in Fourier domain [16], [50]. The family of curvelet functions forms a tight frame, thus, each function $\boldsymbol{u}$ has a representation as

$$\boldsymbol{u} = \sum_{j,l,\mathbf{k}} \langle \boldsymbol{u}, \Psi_{j,l,\mathbf{k}} \rangle \Psi_{j,l,\mathbf{k}}, \tag{6}$$

where $\langle \boldsymbol{u}, \Psi_{j,l,\mathbf{k}} \rangle$ denotes the scalar product of $\boldsymbol{u}$ and $\Psi_{j,l,\mathbf{k}}$. The coefficients $\vartheta_{j,l,\mathbf{k}} = \langle \boldsymbol{u}, \Psi_{j,l,\mathbf{k}} \rangle$ are called curvelet coefficients of function $\boldsymbol{u}$. In matrix notation we can rewrite it as $\boldsymbol{\vartheta} = \boldsymbol{\Psi}\boldsymbol{u}$, and $\boldsymbol{u} = \boldsymbol{\Psi}^{\dagger}\boldsymbol{\vartheta}$, where $\boldsymbol{\Psi}$ is the forward curvelet transform and $\boldsymbol{\Psi}^{\dagger} = (\boldsymbol{\Psi}^T\boldsymbol{\Psi})^{-1}\boldsymbol{\Psi}^T$ is its Moore-Penrose pseudo-inverse transform (corresponding to the minimal dual synthesis frame). Due to the Parseval tight frame property of curvelet transform (i.e., $\boldsymbol{\Psi}^T\boldsymbol{\Psi} = \boldsymbol{I}_n$), we have $\boldsymbol{\Psi}^{\dagger} = \boldsymbol{\Psi}^T$.

C. CS Recovery via Iterative Curvelet Thresholding

As mentioned previously, in [17], we have proposed a new technique for block-based CS image recovery via ICT, namely BCS-ICT. Toward acquiring CS image data, BCS-ICT, firstly, divides each image, of size $\sqrt{n} \times \sqrt{n}$, into small non-overlapping blocks of equal sizes (i.e., size $B \times B$), and then the same sensing matrix $\boldsymbol{\Phi}_B$ (i.e., size $m_B \times B^2$, where $m_B = \lfloor \frac{m \times B^2}{n} \rfloor$) is applied for sampling of each block. In this case, we have

$$\boldsymbol{f}_i = \boldsymbol{\Phi}_B \boldsymbol{u}_i, \tag{7}$$

where $\boldsymbol{u}_i$ is a (column) vector representing block i of the input image, $\boldsymbol{f}_i$ is its corresponding measurement vector and $\boldsymbol{\Phi}_B$ independently samples blocks within an image. Using this technique has several benefits comparing to use of a random sampling operator to the entire image; i.e., the encoder does not need to wait until the entire image is measured, but each block is sent after its linear projection. In addition, at the decoder side, each block is processed independently; therefore the speed of encoding and decoding procedure is increased. Also in this case, we just need to store a $m_B \times B^2$ sensing matrix instead of a $m \times n$ sensing matrix. More precisely, the global sensing matrix takes a block-diagonal structure, $\boldsymbol{\Phi} = \mathrm{diag}(\boldsymbol{\Phi}_B, \cdots, \boldsymbol{\Phi}_B)$, where $\mathrm{diag}(\cdot)$ represents a diagonal matrix.

In order to recover CS images, BCS-ICT considers (2) with the regularization term $\mathcal{R}(\boldsymbol{u}) = \|\boldsymbol{\Psi}\boldsymbol{u}\|_{\ell_1}$ and DCuT [50] as the sparsifying transform. Toward solving this sparsity-promoting ℓ_1 minimization problem, ICT based on projected Landweber [53] iteration is employed. Also, a hard shrinkage function is adapted to provide sparsity-enforcing thresholding while preserving edges and features. In the beginning of each iteration, an adaptive spatial smoothing filter is applied to efficiently reduce the blocking and element like artifacts. We found it beneficial to employ a 3×3 median filter as the smoothing filter, for low measurement rates (below $\%25$). However, for higher measurement ratios (MRs), a Wiener filter the same size is used. Moreover, in [17], we combined two well-known presented IST techniques by Bioucas-Dias and Figueiredo's two-step IST (TwIST) [7] and fast IST algorithm (FISTA) by Beck and Teboulle [6] within the framework of the proposed BCS-ICT to accelerate the execution time. We referred to deploy of both TwIST and FISTA within the BCS-ICT framework as BCS-TwICT and BCS-FICTA, respectively. A detailed description of BCS-FICTA (as a typical case of BCS-ICT) is depicted in Table II.

Evidently, the steps 3 to 6 in Table II can be considered as denoising/shrinkage in curvelet transform domain. In spite of its good performance in removal of the insignificant curvelet coefficients, for enforcing sparsity, it has some drawbacks, e.g., not only is its convergence so slow, but also the linear decay threshold used in the step 4 is highly dependent on the initial thresholding value σ_0 and the (total) number of iterations. Such these parameters are obtained by an empirical-based approach and are not self-adaptive. Delving deeper into this issue reveals that, when the threshold is chosen too large, consequently the number of nonzero-elements is small and the directional sparsity is high, which leads the

TABLE II: BCS-FICTA ALGORITHM

Input: $\boldsymbol{f}$, $\boldsymbol{\Phi}_B$, σ_0, B: block size, k_{max}: maximum iteration number, TOL: tolerance.
Output: $\boldsymbol{u}$: Reconstructed image;
Initialization: Set $k = 0$; $\boldsymbol{u}^{-1} = \boldsymbol{u}^0$; $t^0 = 1$; $\boldsymbol{u}_i^0 = \boldsymbol{\Phi}_B^T \boldsymbol{f}_i$
repeat
1. $\boldsymbol{u}^k = \text{smoothing filter}(\boldsymbol{u}^k)$;
2. **for** each block i **do**
   $\hat{\boldsymbol{u}}_i^k \leftarrow \boldsymbol{u}_i^k + (\frac{t^k-1}{t^{k+1}})(\boldsymbol{u}_i^k - \boldsymbol{u}_i^{k-1})$
   $\hat{\hat{\boldsymbol{u}}}_i^k \leftarrow \hat{\boldsymbol{u}}_i^k + \boldsymbol{\Phi}_B^T(\boldsymbol{f}_i - \boldsymbol{\Phi}_B \hat{\boldsymbol{u}}_i^k)$
   end for
3. $\hat{\boldsymbol{\vartheta}}^k \leftarrow \boldsymbol{\Psi}\hat{\hat{\boldsymbol{u}}}^k$
4. $\sigma^k \leftarrow \sigma_0(1 - k/k_{max})$
5. $\boldsymbol{\vartheta}^k \leftarrow \mathcal{H}_{\boldsymbol{\sigma}^k}(\hat{\boldsymbol{\vartheta}}^k)$
6. $\tilde{\boldsymbol{u}}^k \leftarrow \boldsymbol{\Psi}^T \boldsymbol{\vartheta}^k$
7. **for** each block i **do**
   $\tilde{\boldsymbol{u}}_i^{k+1} \leftarrow \tilde{\boldsymbol{u}}_i^k + \boldsymbol{\Phi}_B^T(\boldsymbol{f}_i - \boldsymbol{\Phi}_B \tilde{\boldsymbol{u}}_i^k)$
   end for
8. Compute $D^{k+1} = \frac{1}{\sqrt{n}} \|\tilde{\boldsymbol{u}}^{k+1} - \hat{\hat{\boldsymbol{u}}}^k\|_{\ell_2}$
9. Update $\boldsymbol{u}^k \leftarrow \tilde{\boldsymbol{u}}^{k+1}$
10. $k \leftarrow k+1$

until $|D^{k+1} - D^k| < \text{TOL}$ or $k = k_{max}$

Note: $\boldsymbol{u}_i^k$ denotes ith block of $\boldsymbol{u}^k$ in kth iteration, by size $B \times B$. $\mathcal{H}_\rho(y) := y \cdot \text{sgn}(|y| - \rho)$ denotes hard shrinkage function, and $\text{sgn}(y) := \{y/|y| \ : \ \forall y \neq 0\} \cup \{0 : \ y = 0\}$ is the sign function.

poor recovery of some edge details. Conversely, unwanted artifacts may be produced if the threshold is too small. In next section, an adaptive curvelet thresholding (ACT) scheme is proposed, which can be incorporated effectively into BCS-ICT, to improve its efficiency.

III. ADAPTIVE CURVELET THRESHOLDING (ACT)

The curvelet transform has been widely used for image denoising because curvelet reconstructions exhibit higher perceptual quality than wavelet based reconstructions, offering visually sharper images and in particular higher quality recovery of edges and of indistinct linear and curve linear features. Almost, all attempts that have been done for denoising in curvelet domain assume that the noisy image is corrupted either by additive white Gaussian noise (AWGN) (e.g., [54], [55]), or by Poisson noise (e.g., [56], [57]). However, in the denoising/shrinkage step indicated in Table II, we assume that the available image $\hat{\hat{\boldsymbol{u}}}^k$ [1] is perturbed and modeled that by additive noise, with no obligation on being AWGN or Poisson. To the best of our knowledge, it is the first time that this paradigm of statistical modeling of perturbations using DCuT is used for CS image restoration. To elucidate on, this step removes insignificant curvelet coefficients (perturbations of available image) and imposes sparsity. Furthermore, this modeling of perturbations by additive noise helps us to model the noise probability density function (PDF) effectively and being able to solve the ensuing optimization problems efficiently. Due to the notations used in the steps 3 to 6 in Table II, the

[1] $\hat{\hat{\boldsymbol{u}}}^k$ is achieved by the steps 1 and 2 of the algorithm shown in Table II, in kth iteration. To avoid confusion, the superscript k is omitted without any confusion, in the subsequent formulas of this section.

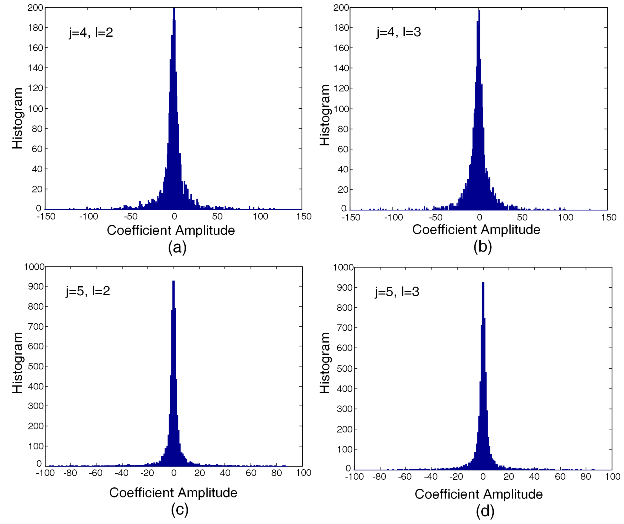


Fig. 2: Marginal distributions of two orientations at two successive scales of the curvelet coefficients of the clean image “Barbara” (512×512).

(available) image coefficients in curvelet domain ($\hat{\boldsymbol{\vartheta}} = \boldsymbol{\Psi}\hat{\hat{\boldsymbol{u}}}$) can be written as

$$\hat{\vartheta}_{j,l,\mathbf{k}} = \vartheta_{j,l,\mathbf{k}} + n_{j,l,\mathbf{k}}, \tag{8}$$

where $\hat{\boldsymbol{\vartheta}}$ and $\boldsymbol{\vartheta}$ are available rough coefficients (noisy) and noise-free (or clean) curvelet coefficients, respectively, and $\boldsymbol{n}$ is unknown additive noise (or *perturbation*) vector. The problem of image denoising in curvelet domain can be expressed as estimation of clean coefficients from noisy data with Bayesian estimation techniques. Either minimum mean-squared error (MMSE) or maximum a posteriori (MAP) estimator is used for solving this problem, the solution requires a prior knowledge about the distribution of curvelet coefficients. More specifically, these methods are optimized w.r.t. the marginal statistics of the coefficients within each sub-band, by imposing a prior distribution on the clean transform coefficients. In this case, a particular success is exhibited by the local approach which considers one (or more) local parameter(s) to model the prior distribution. In other words, local parameters can characterize the intrascale dependency between spatial adjacent. In the local approach, shrinkage functions are data dependent as the value of the neighboring coefficients are used to estimate the local parameters [55].

Fig. 2 shows the histograms of two orientations ($l = 2, 3$) at two successive scales ($j = 4, 5$) of the curvelet coefficients of the clean image “Barbara” (512×512). Obviously, it can be observed that the distributions are characterized by a very sharp peak at zero amplitude and extended tails to both sides of the peak (*leptokurtic distribution*). This leptokurtic property is observed in all histograms of each scale and orientation of all test images. This implies that the curvelet transform is very sparse, since the majority of coefficients have amplitude close to zero [58]. Thus, the marginal statistics of clean curvelet coefficients can be modeled by *generalized Gaussian distribution* (GGD) as a proper distribution to model this leptokurtic behavior. The probability density function (PDF)

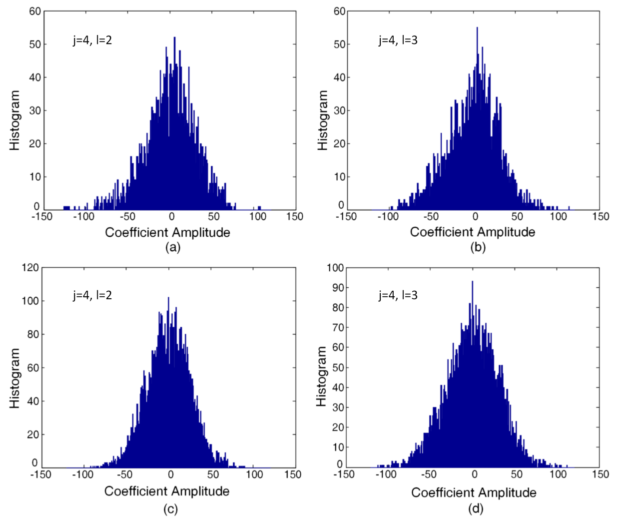


Fig. 3: Histograms of two orientations at two successive scales of the curvelet coefficients of the available image $\hat{\hat{\boldsymbol{u}}}^1$ (where, $\hat{\hat{\boldsymbol{u}}}^1$ is just reconstructed using 0.3 measurements of image "Barbara" by the steps 1 and 2 of BCS-FICTA, in the first iteration).

of GGD is defined as

$$\Pr_{_{GGD}}(\boldsymbol{\chi};\alpha,\beta)=\frac{\beta}{2\alpha\cdot\Gamma(\frac{1}{\beta})}\cdot\exp\big\{-|\frac{\boldsymbol{\chi}}{\alpha}|^{\beta}\big\},\qquad(9)$$

where $\Gamma(\cdot)$ is the Gamma function, i.e., $\Gamma(z)=\int_0^\infty e^{-t}t^{z-1}dt$. Also, $\alpha>0$ indicates the scale parameter which reflects the width of the curve (standard deviation) while $\beta>0$ is called the shape parameter since it determines the degree of sharpness of a GGD curve. In general, α can be derived from β, i.e., $\alpha=\sqrt{\Gamma(1/\beta)\Gamma(3/\beta)}$ in accordance with $\mathrm{E}\{\boldsymbol{\chi}^2\}=1$, where $\mathrm{E}\{\cdot\}$ denotes the mathematical expectation. The GGD model contains the Laplacian and Gaussian PDFs as a special cases, using $\beta=1$ and $\beta=2$, respectively. Also, if $0<\beta<1$, then $\Pr_{_{GGD}}(\chi)$ specifies the hyper-Laplacian distribution. In this paper, Laplacian distribution is chosen to model the marginal distributions of curvelet coefficients of the clean image, i.e.,

$$\Pr_{\boldsymbol{\vartheta}}(\vartheta_{j,l,\mathbf{k}})=\frac{1}{\sqrt{2}\sigma_{j,l,\mathbf{k}}}\cdot\exp\big(\frac{-\sqrt{2}\;|\vartheta_{j,l,\mathbf{k}}|}{\sigma_{j,l,\mathbf{k}}}\big),\qquad(10)$$

where $\sigma_{j,l,\mathbf{k}}$ is the standard deviation of clean coefficients. This choice is despite the fact that marginal distributions of curvelet coefficients have heavier tails than a Laplacian distribution; tend to go zero slower than a Gaussian distribution; and can be well modelled by a hyper-Laplacian distribution. But, actually, this choice makes a trade-off between modeling the coefficients statistics accurately and being able to efficiently attack the ensuing optimization problem.

The histograms of two orientations ($l=2,3$) at two successive scales ($j=4,5$) of the curvelet coefficients of the initial available image $\hat{\hat{\boldsymbol{u}}}^1$ are shown in Fig. 3 (corresponding to the histograms of Fig. 2). More specifically, these histograms are obtained using the curvelet coefficients of the available image $\hat{\hat{\boldsymbol{u}}}$ which is achieved by the steps1 and 2 of BCS-FICTA algorithm, in the first iteration (see Table II). As can be seen, here, the leptokurtic behavior and heavy tails do not exist in the shown distributions. More accurately, these two properties do not exist in all distributions of each scale and orientation of all available images achieved by the steps 1 and 2.

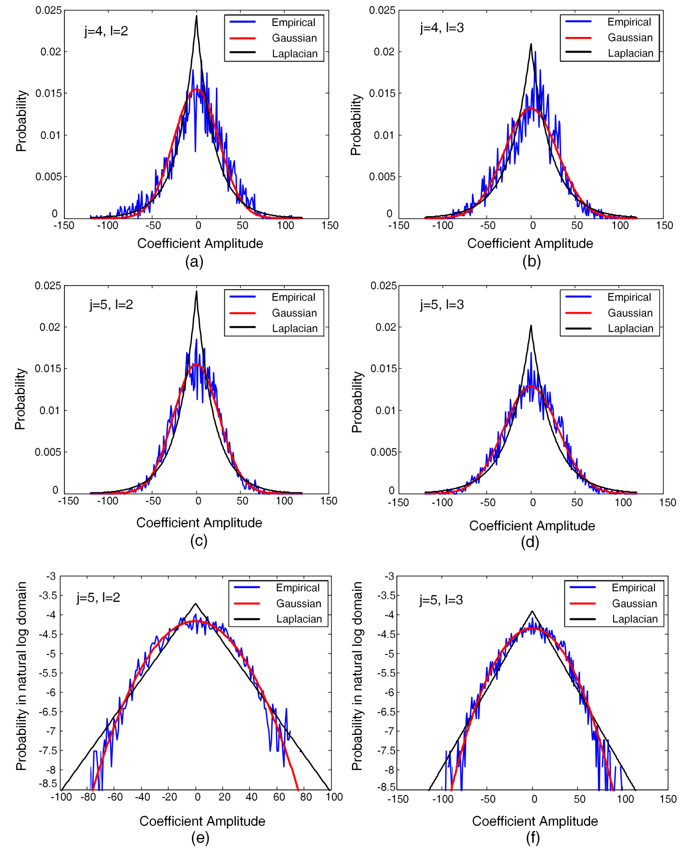


Fig. 4: (a)-(d) PDFs of noise distribution (empirical perturbations) at two orientations of two successive scales (corresponding to the histograms of Figs. 2 and 3) along with their best fitted PDFs: the noise distribution (blue), the Gaussian fit (red), and the Laplacian fit (black). (e) and (f) show the same distributions of (c) and (d) in natural log domain.

Considering (8), the noise distribution can be obtained using the curvelet coefficients of the clean image and the available image. Fig. 4 shows the PDFs of noise distribution at two orientations of two successive scales corresponding to those of Figs. 2 and 3. It can be observed that the empirical distributions of noise coefficients $n_{j,l,\mathbf{k}}$ can be well characterized by Gaussian distributions, while the Laplacian distributions have much larger fitting errors. This observation motivates us to model the PDF of $n_{j,l,\mathbf{k}}$ by a standard Gaussian distribution, i.e.,

$$\Pr_{\boldsymbol{n}}(n_{j,l,\mathbf{k}})=\frac{1}{\sqrt{2\pi}\sigma_{n_{j,l}}}\cdot\exp\big(\frac{-n^2_{j,l,\mathbf{k}}}{2\sigma^2_{n_{j,l}}}\big),\qquad(11)$$

where $\sigma^2_{n_{j,l}}$ is the noise variance in curvelet domain, which can be estimated using robust median estimator [59] and Monte-Carlo simulations [50].

Consider the marginal distribution of the clean curvelet coefficients (e.g., see (10)). Since in CS recovery problems, we do not have access to the clean image, the variance of the clean image curvelet coefficients $\sigma^2_{j,l,\mathbf{k}}$ can be estimated by *maximum likelihood* (ML) estimator, through averaging over

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

the neighboring noisy (available $\hat{\vartheta}_{j,l,\mathbf{k}}$) coefficients within a squared window $\mathcal{N}(\mathbf{k})$ centered at $\mathbf{k}$:

$$\begin{aligned}\sigma^2_{j,l,\mathbf{k}} &= \arg\max_{\sigma^2_{j,l,\mathbf{k}}} \prod_{\mathbf{k}'\in\mathcal{N}(\mathbf{k})} \mathrm{Pr}_{\hat{\boldsymbol{\vartheta}}|\boldsymbol{\sigma}}(\hat{\vartheta}_{j,l,\mathbf{k}'}|\sigma^2_{j,l,\mathbf{k}}) \\ &= \max\big(0, \frac{1}{\mathcal{M}} \sum_{\mathbf{k}'\in\mathcal{N}(\mathbf{k})} \hat{\vartheta}^2_{j,l,\mathbf{k}'} - \sigma^2_{n_{j,l}}\big), \qquad (12)\end{aligned}$$

where $\mathrm{Pr}_{\hat{\boldsymbol{\vartheta}}|\boldsymbol{\sigma}}\left(\cdot|\sigma^2_{j,l,\mathbf{k}}\right)$ is the Gaussian distribution with zero mean and variance $\sigma^2_{j,l,\mathbf{k}} + \sigma^2_{n_{j,l}}$, and $\mathcal{M}$ is the number of coefficients in $\mathcal{N}(\mathbf{k})$ [60].

In order to estimate $\vartheta_{j,l,\mathbf{k}}$ from the available observation $\hat{\vartheta}_{j,l,\mathbf{k}}$, the MAP estimator is employed, i.e.,

$$\vartheta_{j,l,\mathbf{k}} = \arg\max_{\vartheta_{j,l,\mathbf{k}}} \left\{ \mathrm{Pr}_{\boldsymbol{\vartheta}|\hat{\boldsymbol{\vartheta}}}\left(\vartheta_{j,l,\mathbf{k}}|\hat{\vartheta}_{j,l,\mathbf{k}}\right)\right\}. \qquad (13)$$

By considering (8) and using Bayes' rule, we get

$$\begin{aligned}\vartheta_{j,l,\mathbf{k}} &= \arg\max_{\vartheta_{j,l,\mathbf{k}}} \left\{ \mathrm{Pr}_{\hat{\boldsymbol{\vartheta}}|\boldsymbol{\vartheta}}\left(\hat{\vartheta}_{j,l,\mathbf{k}}|\vartheta_{j,l,\mathbf{k}}\right) \cdot \mathrm{Pr}_{\boldsymbol{\vartheta}}\left(\vartheta_{j,l,\mathbf{k}}\right)\right\} \\ &= \arg\max_{\vartheta_{j,l,\mathbf{k}}} \left\{ \mathrm{Pr}_{\boldsymbol{n}}\left(\hat{\vartheta}_{j,l,\mathbf{k}} - \vartheta_{j,l,\mathbf{k}}\right) \cdot \mathrm{Pr}_{\boldsymbol{\vartheta}}\left(\vartheta_{j,l,\mathbf{k}}\right)\right\}. \qquad (14)\end{aligned}$$

Therefore, (14) allows us to write this estimation in terms of the PDF of the noise coefficient ($\mathrm{Pr}_{\boldsymbol{n}}$), and the PDF of the clean image coefficient ($\mathrm{Pr}_{\boldsymbol{\vartheta}}$). By substituting (10) and (11) into (14) and doing some manipulations, we obtain

$$\vartheta_{j,l,\mathbf{k}} = \arg\min_{\vartheta_{j,l,\mathbf{k}}} \left\{ \frac{1}{2}\left|\hat{\vartheta}_{j,l,\mathbf{k}} - \vartheta_{j,l,\mathbf{k}}\right|^2 + \frac{\sqrt{2}\sigma^2_{n_{j,l}}}{\sigma_{j,l,\mathbf{k}}}\left|\vartheta_{j,l,\mathbf{k}}\right| \right\}. \qquad (15)$$

Hence, the estimation of $\vartheta_{j,l,\mathbf{k}}$ is achieved as

$$\vartheta_{j,l,\mathbf{k}} = \mathrm{sgn}(\hat{\vartheta}_{j,l,\mathbf{k}}) \max\big(0, |\hat{\vartheta}_{j,l,\mathbf{k}}| - \frac{\sqrt{2}\sigma^2_{n_{j,l}}}{\sigma_{j,l,\mathbf{k}}}\big), \qquad (16)$$

where, in fact, (16) is the classical soft shrinkage function [61], defined as $\mathcal{S}_\rho(y) := \mathrm{sgn}(y)\cdot\max(0, |y| - \rho)$.

As mentioned before, one of the contributions of this paper is introducing a denoising/shrinkage method, namely ACT, trying to adaptively remove the perturbations appeared in recovered images during the CS recovering process, by removing insignificant curvelet coefficients, and imposing sparsity. Also, ACT is self-adaptive and more efficient than the empirical-based linear decay thresholding used in [17]. Employing ACT within the frameworks of BCS-FICTA would promote its efficiency and lead to a simple and effective CS image recovery method (see Fig. 5). Although its results are still far from those obtained by the state-of-the-art techniques in CS image recovery, its incorporation into the proposed JASR would result in an efficient CS image recovery scheme that surpasses all of them with notably large margins (see Section VI-A). It is worth noting that in [66], we have extended the proposed ACT for mixed Gaussian-Impulse noise removal problem which achieves noteworthy results.

IV. JOINT ADAPTIVE SPARSITY REGULARIZATION (JASR)

As stated previously, using only the local sparsity constraint in (2) may not lead to an enough accurate CS image restoration. An alternative approach for superior incorporating the

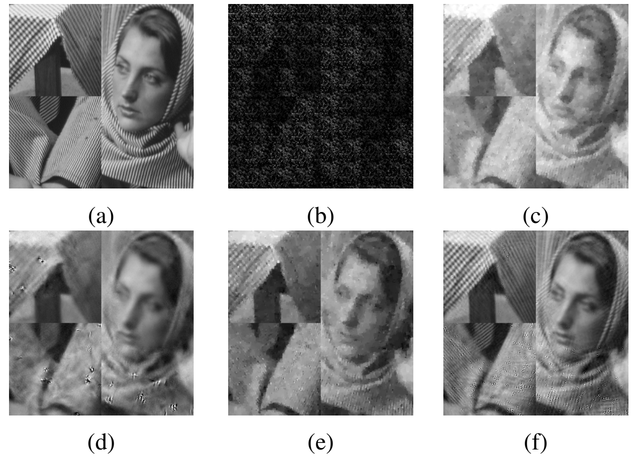


Fig. 5: Visual quality comparison of CS recovered *Barbara* (512×512) images with 10% of measurements. (a) Original image. The CS recovered images by: (b) direct recovery using the steps 1 and 2 of the BCS-FICTA algorithm, in the first iteration, i.e., $\hat{\boldsymbol{u}}^1$ (PSNR=12.58 dB, SSIM=0.030); (c) TV via augmented Lagrangian method (TVAL3) [62] (PSNR=18.93 dB, SSIM=0.537); (d) block-based CS image recovery using smoothed projected Landweber iterations (BCS-SPL) [63] (PSNR=22.59 dB, SSIM=0.687); (e) nonparametric Bayesian CS using beta process analysis model (we refer to it as BPFA-CS) [64] (PSNR=22.94 dB, SSIM=0.579); and (f) BCS-FICTA [17] via the proposed ACT (PSNR=23.67 dB, SSIM=0.724). Cropped portions of the original and the reconstructed images are shown.

prior knowledge about images is via sparse representation and nonlocal self-similarity which has led to highly competent works of sparsity-based CS image restoration (see e.g., [44] and [48]). As can be seen in Fig. 1, the distribution of transform coefficients $\boldsymbol{\Theta_u}$ is characterized by a very sharp peak at zero amplitude and the vast majority of coefficients are concentrated near zero (*leptokurtic distribution*) which implies that $\boldsymbol{\Theta_u}$ has a very sparse representation. Considering the above point, NLSM for self-similarity in 3D transform domain motivated us to adopt it, by some modifications, as the adaptive *nonlocal statistical sparsity measure* (NSSM) in our proposed scheme to depict the repetitiveness of the textures and structures in natural images within nonlocal regions. The major similarities and differences between NSSM and those mentioned in Section II-A will be discussed in the following.

The proposed *joint adaptive sparsity regularization* (JASR) is defined by integrating both the local sparsity constraint (which depicts the local smoothness and geometric regularity of image structures achieved by DCuT) and the NSSM in 3D transform domain (corresponding to the nonlocal self-similarity prior achieved by a method similar to the ones introduced in Section II-A, that is NSSM). In a mathematical expression, the proposed JASR prior is expressed as

$$\begin{aligned}\mathcal{R}_{\mathrm{JASR}}(\boldsymbol{u}) &= \Psi_{\mathrm{DCuT}}(\boldsymbol{u}) + \tau\Psi_{\mathrm{NSSM}}(\boldsymbol{u}) \\ &= \|\boldsymbol{\Psi u}\|_{\ell_1} + \tau\|\boldsymbol{\Theta_u}\|_{\ell_1}, \qquad (17)\end{aligned}$$

where the first and the second terms represent the image local smoothness prior and nonlocal self-similarity prior, respectively. Also, τ is a regularization parameter which controls the trade-off between two competing (statistical) sparsity terms.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

The similarities and differences between the proposed JASR, and those proposed in [20] (JSM) and [48] (CoSM) are outlined as the following.

- Resemble to NLSM, the proposed NSSM characterizes the nonlocal self-similarity by means of the distributions of the transform coefficients, statistical modeling, while the nonlocal self-similarity in CoSM (N3D) is characterized by means of the sparsity of transform coefficients—summing all the number of nonzero coefficients of each group.
- In both NLSM and N3D, the 3D transform denoted by $\mathcal{T}^{3D}$ is composed of 2D DCT and 1D wavelet transform, however in NSSM in the proposed JASR, $\mathcal{T}^{3D}$ is composed of 3D wavelet transform. More specifically, on the 2D wavelet coefficients of each patch in a group $\boldsymbol{G}_{\boldsymbol{u}_{p_l{}'}}$, 1D wavelet transform is conducted along the third axis.
- CoSM is developed in order to recover CS images; however, JSM is incorporated into the regularization-based framework for the image inverse problems such as image deblurring, image inpainting and mixed Gaussian plus impulse noise removal, except CS. In this paper, though the proposed JASR is developed for CS image restoration, it can be used appropriately for the other image inverse problems like those mentioned for JSM.
- RCoS and JSM, both employ an anisotropic TV regularization to characterize the local smoothness prior of natural images. Nevertheless, JASR takes the advantages of local sparsity achieved by DCuT for better characterizing of the local smoothness prior and also for better edge and detail preserving.

V. CS Image Recovery via JASR

By substituting the proposed JASR (17) for the regularization term $\mathcal{R}(\boldsymbol{u})$ in the regularization-based framework of (2), the proposed optimization problem for CS image recovery is expressed as follows:

$$\min_{\boldsymbol{u}}\{\frac{1}{2}\|\boldsymbol{f}-\boldsymbol{\Phi u}\|_{\ell_2}^2+\lambda\|\boldsymbol{\Psi u}\|_{\ell_1}+\tau'\|\boldsymbol{\Theta_u}\|_{\ell_1}\}. \tag{18}$$

The first term represents the observation constraint and the second and the third term indicate the image local smoothness and nonlocal self-similarity prior information.

In order to solve the above optimization problem efficiently, the framework of the split Bregman iteration (SBI) algorithm [8] is employed; owing to its efficiencies in efficiently solving the large-scale optimization problems, small memory footprint requirement, and also, ease of programming by users. For this purpose, first, (18) is converted into its equivalent constrained-form, and then the variable splitting technique is introduced along with invoking the Bregman iteration.

By introducing two auxiliary variables $\boldsymbol{\vartheta}$ and $\boldsymbol{w}$, (18) can be converted into its corresponding equivalent constrained-form:

$$\begin{aligned}&\min_{\boldsymbol{u},\boldsymbol{\vartheta},\boldsymbol{w}}\{\frac{1}{2}\|\boldsymbol{f}-\boldsymbol{\Phi u}\|_{\ell_2}^2+\lambda\|\boldsymbol{\vartheta}\|_{\ell_1}+\tau'\|\boldsymbol{\Theta_w}\|_{\ell_1}\}\\&s.t.\quad \boldsymbol{\vartheta}=\boldsymbol{\Psi u},\ \boldsymbol{u}=\boldsymbol{w},\ \boldsymbol{w}=\Omega_{\text{NSSM}}(\boldsymbol{\Theta_w}).\end{aligned} \tag{19}$$

In order to solve the above minimization problem, an alternating SBI algorithm is applied. We finally obtain the following schemes:

$$\begin{aligned}\boldsymbol{u}^{k+1}=&\arg\min_{\boldsymbol{u}}\frac{1}{2}\|\boldsymbol{f}-\boldsymbol{\Phi u}\|_{\ell_2}^2+\frac{\mu_1}{2}\|\boldsymbol{\vartheta}^k-\boldsymbol{\Psi u}-b^k\|_{\ell_2}^2\\&+\frac{\mu_2}{2}\|\boldsymbol{u}-\boldsymbol{w}^k-\boldsymbol{c}^k\|_{\ell_2}^2,\end{aligned} \tag{20a}$$

$$\boldsymbol{\vartheta}^{k+1}=\arg\min_{\boldsymbol{\vartheta}}\lambda\|\boldsymbol{\vartheta}\|_{\ell_1}+\frac{\mu_1}{2}\|\boldsymbol{\vartheta}-\boldsymbol{\Psi u}^{k+1}-\boldsymbol{b}^k\|_{\ell_2}^2, \tag{20b}$$

$$\boldsymbol{w}^{k+1}=\arg\min_{\boldsymbol{w}}\tau'\|\boldsymbol{\Theta_w}\|_{\ell_1}+\frac{\mu_2}{2}\|\boldsymbol{u}^{k+1}-\boldsymbol{w}-\boldsymbol{c}^k\|_{\ell_2}^2, \tag{20c}$$

$$\boldsymbol{b}^{k+1}=\boldsymbol{b}^k-\boldsymbol{\vartheta}^{k+1}+\boldsymbol{\Psi u}^{k+1}, \tag{20d}$$

$$\boldsymbol{c}^{k+1}=\boldsymbol{c}^k-\boldsymbol{u}^{k+1}+\boldsymbol{w}^{k+1}. \tag{20e}$$

Here, μ_1 and μ_2 are fixed value parameters for improving the numerical stability of the algorithm.

Given $\boldsymbol{\vartheta}^k$ and $\boldsymbol{w}^k$, the sub-problem of (20a) consists of minimizing a strictly convex quadratic function that can be solved easily. By setting the gradient of the objective function in (20a) to zero, a closed-form solution for (20a) is achieved:

$$\begin{aligned}\boldsymbol{u}^{k+1}=&\big(\boldsymbol{\Phi}^T\boldsymbol{\Phi}+(\mu_1+\mu_2)\boldsymbol{I}_n\big)^{-1}\Big(\mu_1\boldsymbol{\Psi}^T(\boldsymbol{\vartheta}^k-\boldsymbol{b}^k)\\&+\mu_2(\boldsymbol{w}^k+\boldsymbol{c}^k)+\boldsymbol{\Phi}^T\boldsymbol{f}\Big).\end{aligned} \tag{21}$$

Since for image CS recovery, $\boldsymbol{\Phi}$ is a random projection matrix with no especial structures ($\boldsymbol{\Phi}^T\boldsymbol{\Phi}\neq\boldsymbol{I}_n$), it is too costly to solve the minimization of the quadratic function in (20a) directly by using (21), because of existence of the matrix inverse in (21). Here, in order to avoid computing the inversion of matrix, the steepest descent method with the optimal step size is utilized to solve the minimization of the quadratic function in (20a), which can be expressed as

$$\boldsymbol{u}^{k+1}=\boldsymbol{u}^k-\eta^k\boldsymbol{g}^k,\quad \eta^k>0. \tag{22}$$

Here, $\boldsymbol{g}$ is the gradient direction of the objective function and $\eta=\text{abs}\Big(\boldsymbol{g}^T\boldsymbol{g}/\boldsymbol{g}^T\big(\boldsymbol{\Phi}^T\boldsymbol{\Phi}+(\mu_1+\mu_2)\boldsymbol{I}_n\big)\boldsymbol{g}\Big)$ represents the optimal step size.

By given $\boldsymbol{u}$ in hand (according to (20a)) and considering $\hat{\boldsymbol{\vartheta}}=\boldsymbol{\Psi u}+\boldsymbol{b}$ (for simplicity, the superscript k is dropped without confusion), the sub-problem of (20b) becomes

$$\min_{\boldsymbol{\vartheta}}\frac{1}{2}\|\boldsymbol{\vartheta}-\hat{\boldsymbol{\vartheta}}\|_{\ell_2}^2+\frac{\lambda}{\mu_1}\|\boldsymbol{\vartheta}\|_{\ell_1}. \tag{23}$$

By these transformations, we regard $\boldsymbol{\vartheta}'$ as some type of the noisy unknown coefficient $\boldsymbol{\vartheta}$. More precisely, the sub-problem of (23) can be interpreted as the denoising in curvelet coefficient domain. Considering the fact that the unkonwn variable $\boldsymbol{\vartheta}$ is component-wise separable in curvelet domain, each of its component $\vartheta_{j,l,\mathbf{k}}$ can be obtained by a component-wise shrinkage procedure, i.e.,

$$\vartheta_{j,l,\mathbf{k}}=\arg\min_{\vartheta_{j,l,\mathbf{k}}}\left\{\frac{1}{2}\left|\vartheta_{j,l,\mathbf{k}}-\hat{\vartheta}_{j,l,\mathbf{k}}\right|^2+\frac{\lambda_{j,l,\mathbf{k}}}{\mu_1}\left|\vartheta_{j,l,\mathbf{k}}\right|\right\}. \tag{24}$$

Considering (24) and the one introduced in (15) with regarding $\frac{\lambda_{j,l,\mathbf{k}}}{\mu_1}=\frac{\sqrt{2}\sigma_{n_{j,l}}^2}{\sigma_{j,l,\mathbf{k}}}$, the $\boldsymbol{\vartheta}$-sub-problem of (23) can be interpreted as the proposed ACT introduced in Section III. Therefore, the parameter λ is self-adaptive.

Given $\boldsymbol{u}$, analogously, the sub-problem of (20c) can be written as

$$\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w}-\boldsymbol{z}\|_{\ell_2}^2 + \frac{\tau'}{\mu_2}\|\boldsymbol{\Theta_w}\|_{\ell_1}, \tag{25}$$

where $\boldsymbol{z} = \boldsymbol{u} - \boldsymbol{c}$ (for simplicity the superscript k is omitted without confusion). Due to the complicated definition of $\boldsymbol{\Theta_w}$, it seems difficult to solve (25) directly. In order to solve (25) amenable, in this paper, a reasonable assumption is used which leads to obtain a closed-form solution of (25). Such this assumption is efficiently employed in [20] and [48]; hence, we utilize the same approach to solve the above minimization problem. Accordingly, by viewing $\boldsymbol{z}$ as some type of the noisy observation $\boldsymbol{w}$, we denote the error vector by $\boldsymbol{e}' = \boldsymbol{w}-\boldsymbol{z}$ (such that $\boldsymbol{w}, \boldsymbol{z} \in \mathbb{R}^n$) and each element of $\boldsymbol{e}'$ by $\boldsymbol{e}'(i), i = 1, \cdots, n$. Then an assumption is made that all elements of $\boldsymbol{e}'$ are independent and identically distributed (i.i.d.) with zero-mean and variance ${\sigma'}^2$. It is all transparent that the above assumption does not need to be Gaussian, Laplacian or GGD process. Based on the assumption, and by invoking the *law of large numbers* in probability theory, for any $\epsilon' > 0$, it leads to $\lim_{n\to\infty} \Pr\{|\frac{1}{n}\sum_{i=1}^n {\boldsymbol{e}'}^2(i) - {\sigma'}^2| < \frac{\epsilon'}{2}\} = 1$; thus, we have

$$\lim_{n\to\infty} \Pr\{\Big|\frac{1}{n}\|\boldsymbol{w}-\boldsymbol{z}\|_{\ell_2}^2 - {\sigma'}^2\Big| < \frac{\epsilon'}{2}\} = 1. \tag{26}$$

Let $\boldsymbol{\Theta_{e'}} = \boldsymbol{\Theta_w} - \boldsymbol{\Theta_z}$ be the error vector in 3D transform domain (such that $\boldsymbol{\Theta_w}, \boldsymbol{\Theta_z} \in \mathbb{R}^{K_\Theta}$), and let $\boldsymbol{\Theta_{e'}}(j'), j' = 1, 2, \cdots, K_\Theta$ denote each element of $\boldsymbol{\Theta_{e'}}$. According to the orthogonal property of the 3D transform $\mathcal{T}^{3D}$ with respect to this fact that $\mathcal{T}^{3D}$ does not alter the variance of each group $\boldsymbol{G}_{\boldsymbol{e}'_{p_{l'}}}$, $l' = 1, 2, \cdots, P$, and due to the definition of the error vector in 3D transform domain, $\boldsymbol{\Theta_{e'}}$, it is concluded that all elements of $\boldsymbol{\Theta_{e'}}$ are i.i.d. with zero-mean and variance ${\sigma'}^2$. Thanks to the *law of large numbers*, by doing the same manipulations with (26) to $\boldsymbol{\Theta_{e'}}(j')$, for any $\epsilon' > 0$, it yields $\lim_{K_\Theta\to\infty} \Pr\{|\frac{1}{K_\Theta}\sum_{j'=1}^{K_\Theta} \boldsymbol{\Theta}^2_{\boldsymbol{e}'}(j') - {\sigma'}^2| < \frac{\epsilon'}{2}\} = 1$; hence, we have

$$\lim_{K_\Theta\to\infty} \Pr\{\Big|\frac{1}{K_\Theta}\|\boldsymbol{\Theta_w}-\boldsymbol{\Theta_z}\|_{\ell_2}^2 - {\sigma'}^2\Big| < \frac{\epsilon'}{2}\} = 1. \tag{27}$$

Therefore, according to (26) and (27), the following property is concluded:

$$\lim_{\substack{n\to\infty \\ K_\Theta\to\infty}} \Pr\{\Big|\frac{1}{n}\|\boldsymbol{w}-\boldsymbol{z}\|_{\ell_2}^2 - \frac{1}{K_\Theta}\|\boldsymbol{\Theta_w}-\boldsymbol{\Theta_z}\|_{\ell_2}^2\Big| < \epsilon'\} = 1, \tag{28}$$

in which the relationship between $\|\boldsymbol{w}-\boldsymbol{z}\|_{\ell_2}^2$ and $\|\boldsymbol{\Theta_w}-\boldsymbol{\Theta_z}\|_{\ell_2}^2$ (almost sure) is described as follows:

$$\frac{1}{n}\|\boldsymbol{w}-\boldsymbol{z}\|_{\ell_2}^2 = \frac{1}{K_\Theta}\|\boldsymbol{\Theta_w}-\boldsymbol{\Theta_z}\|_{\ell_2}^2. \tag{29}$$

Incorporating (29) into (25), leads to

$$\arg\min_{\boldsymbol{\Theta_w}} \frac{1}{2}\|\boldsymbol{\Theta_w}-\boldsymbol{\Theta_z}\|_{\ell_2}^2 + \frac{K_\Theta\tau'}{\mu_2\ n}\|\boldsymbol{\Theta_w}\|_{\ell_1}. \tag{30}$$

Since the unknown variable $\boldsymbol{\Theta_w}$ is component-wise separable in (30), each of its component $\boldsymbol{\Theta_w}(j')$ can be independently obtained by a component-wise (soft) shrinkage procedure, i.e.,

$$\boldsymbol{\Theta_w}(j') = \arg\min_{\boldsymbol{\Theta_w}(j')} \Big\{\frac{1}{2}|\boldsymbol{\Theta_w}(j')-\boldsymbol{\Theta_z}(j')|^2 + \frac{K_\Theta\tau'}{\mu_2 n}|\boldsymbol{\Theta_w}(j')|\Big\}, \tag{31}$$

where $0 \in \frac{\boldsymbol{\Theta_w}(j')}{|\boldsymbol{\Theta_w}(j')|}(\frac{K_\Theta\tau'}{\mu_2 n}) + \big(\boldsymbol{\Theta_w}(j') - \boldsymbol{\Theta_z}(j')\big)$. Indeed, in vector form, this shrinkage can be written shortly as: $\boldsymbol{\Theta_w} = \mathcal{S}_{\frac{K_\Theta\tau'}{\mu_2 n}}\big(\boldsymbol{\Theta_z}\big)$, where $\mathcal{S}_\rho(\boldsymbol{y}) := \operatorname{sgn}(\boldsymbol{y}) \odot \max\big\{|\boldsymbol{y}| - \boldsymbol{\rho}, \boldsymbol{0}\big\}$ is soft shrinkage operator [61] and $\odot$ denotes the component-wise product. Here, the sign, max and absolute value functions are applied in component-wise fashion. Thus, the closed-form solution of (25) is written as

$$\boldsymbol{w} = \Omega_{\text{NSSM}}\big(\boldsymbol{\Theta_w}\big) = \Omega_{\text{NSSM}}\big(\mathcal{S}_{\frac{K_\Theta\tau'}{\mu_2 n}}\big(\boldsymbol{\Theta_z}\big)\big). \tag{32}$$

Exploring nonlocal self-similarity of an image (by stacking the similar image patches into multiple groups and then imposing the correlation prior within each group) requires the original image. Nevertheless, in practice, the original image is unknown in CS, except for its random measurements $\boldsymbol{f}$. Such a problem with a chicken-and-egg flavor is usually solved by an iterative scheme; obtaining both the estimate of the original image and its patch-grouping information in NSSM, alternately. In this paper, we set the initial estimate for the original image by BCS-FICTA [17] via the proposed ACT owing to its simplicity and reasonable efficiencies in both the quality of recovery and the computational time. Also, in the subsequent iterations, since $\boldsymbol{z}$ in (25) is regarded as a good available approximation of the original image, it is conducted to obtain the patch-grouping information for NSSM using all patches extracted from $\boldsymbol{z}$.

The proposed algorithm for CS image recovery via JASR is summarized in detail in Table III.

VI. Experimental Results

In this section, we evaluate the performance of the proposed methods—BCS-FICTA via the proposed ACT and CS-JASR—and compare them with several benchmark methods. The performances of our experiments are evaluated on the gray-scale test images (of size 256×256) shown in Fig. 6. These test images are commonly used in other related publication, which enables a fair comparison of results. In all experiments we use the block size of 32×32 and the CS measurement of each block is obtained by applying a dense Gaussian projection matrix to each of them. All the experimental results reported in this paper are averaged over 10 independent trials. Also, the second generation of DCuT via wrapping [50] is employed in our implementations. All experiments were performed using MATLAB 2012a, on a computer equipped with Intel® core^{TM} i7, 3.5 GHz processor, with 16 GB of RAM, and running on Windows 7. Due to the space limitations, only parts of the experimental results are shown in this paper. Interested readers may contact the corresponding author for all the images and the source code.

A. Comparison with State-of-the-Art Techniques

To demonstrate the effectiveness of the proposed methods in CS image recovery, we have compared each of which with several competitive CS image recovery techniques: (i) TVAL3 [62]; (ii) BCS-SPL [63]; (iii) BPFA-CS [64]; (iv) TV-based CS image recovery via nonlocal regularization (TVNLR) [42]; (v) NLCS [46]; (vi) multi-hypothesis (MH) prediction method

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

TABLE III: THE DETAILED DESCRIPTION OF THE PROPOSED CS IMAGE RECOVERY FRAMEWORK VIA JASR

Input: $\boldsymbol{f}$, $\boldsymbol{\Phi}_B, B_p, \omega, \tau', \mu_1, \mu_2, J_0$: inloop iteration number
k'_{max}: maximum iteration number, Tol: Tolerance
Output: $\boldsymbol{u}^*$: Recovered image;
Initialization: Set $k = 0$; $(\boldsymbol{b}^0, \boldsymbol{c}^0) = (\mathbf{0}, \mathbf{0})$
$\boldsymbol{u}_{init} = \boldsymbol{u}^0$= BCS-FICTA($\boldsymbol{f}$,$\boldsymbol{\Phi}_B$) [17] via the proposed ACT
While a stop criterion is not satisfied **do**
1. $\boldsymbol{u}^{k+1} = \tilde{\boldsymbol{u}} = \boldsymbol{u}^k$
   $\hat{\boldsymbol{\vartheta}}^{k+1} = \boldsymbol{\Psi}\boldsymbol{u}^{k+1} + \boldsymbol{b}^k$
   $\boldsymbol{z}^{k+1} = \boldsymbol{u}^{k+1} - \boldsymbol{c}^k$
2. **for** each element of $\boldsymbol{\vartheta}$ **do**
   compute $\vartheta_{j,l,\mathbf{k}}^{k+1}$ using (24)
   end for
3. update $\boldsymbol{\vartheta}^{k+1}$ by concatenating all $\{\vartheta_{j,l,\mathbf{k}}^{k+1}\}$
4. **for** each patch $\boldsymbol{z}_{p_{l'}}$ **do**
   find $\boldsymbol{G}_{\boldsymbol{z}_{p_{l'}}}$, compute $\mathcal{T}^{3D}(\boldsymbol{G}_{\boldsymbol{z}_{p_{l'}}})$
   end for
5. arrange all $\mathcal{T}^{3D}(\boldsymbol{G}_{\boldsymbol{z}_{p_{l'}}})$ in lexicographic order to form $\boldsymbol{\Theta}_{\boldsymbol{z}}$
6. update $\boldsymbol{w}^{k+1}$ using (32)
7. **for** $i = 1 : J_0$
   $\boldsymbol{g}_i^k \leftarrow \big(\boldsymbol{\Phi}^T(\boldsymbol{\Phi}\boldsymbol{u}^{k+1} - \boldsymbol{f}) + \mu_1\boldsymbol{\Psi}^T(\boldsymbol{\Psi}\boldsymbol{u}^{k+1} - \boldsymbol{\vartheta}^{k+1} + \boldsymbol{b}^k)$
   $+\mu_2(\boldsymbol{u}^{k+1} - \boldsymbol{w}^{k+1} - \boldsymbol{c}^k)\big)$
   $\eta_i^k \leftarrow \mathrm{diag}\big(\mathrm{abs}(\frac{{\boldsymbol{g}_i^k}^T\boldsymbol{g}_i^k}{{\boldsymbol{g}_i^k}^T\left(\boldsymbol{\Phi}^T\boldsymbol{\Phi}+(\mu_1+\mu_2)\boldsymbol{I}_n\right)\boldsymbol{g}_i^k})\big)$
   $\boldsymbol{u}^{k+1} \leftarrow \boldsymbol{u}^{k+1} - \eta_i^k\boldsymbol{g}_i^k$
   end for
8. compute s^{k+1}= SSIM($\boldsymbol{u}^{k+1}, \tilde{\boldsymbol{u}}$)
9. compute *diff* = abs($s^{k+1} - s^k$)
10. update $\boldsymbol{b}^k, \boldsymbol{c}^k$ using (20d) and (20e), respectively
11. $k \leftarrow k + 1$

end While
stopping criterion: $k = k'_{max}$ or *diff* $\leq Tol$.

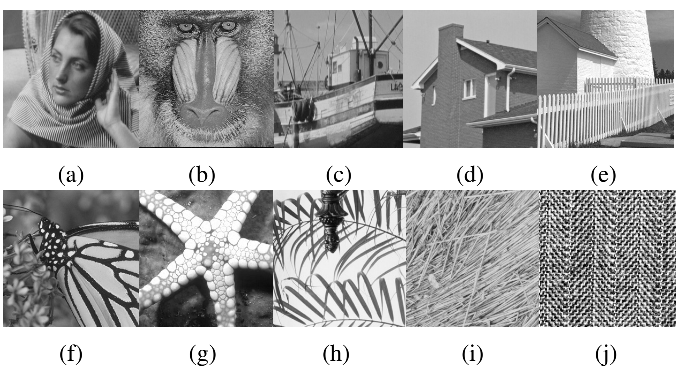


Fig. 6. The test images used in the experiments. (a)-(j): *Barbara*, *Baboon*, *Boat*, *House*, *Fence*, *Monarch*, *Starfish*, *Leaves*, *Straw* and *Texture*.

[65]; (vii) RCoS [48]; (viii) MARX-PC [45]; (ix) CS image recovery using adaptively learned sparsifying basis (we refer to it as CS-ALS) [67]; and (x) CS image restoration based on wavelet sparse representation (denoted as WSR-μ) [68]. It is worth emphasizing that the last six of these methods are known as the current state-of-the-art methods for CS image recovery. The source codes of all benchmark methods of [42], [45]-[48], and [62]-[68] were obtained from their respective authors' websites[2]. Note that, to make a fair comparison

[2]we thank their respective authors for providing the source codes.

TABLE IV: THE AVERAGE PSNR/SSIM COMPARISONS AMONG TVAL3, BPFA-CS, BCS-SPL AND BCS-FICTA VIA THE PROPOSED ACT, PERFORMED ON THE ALL TEST IMAGES (SHOWN IN FIG. 6) WITH DIFFERENT MEASUREMENT RATIOS.

	MR		
Algorithm	0.1	0.3	0.5
TVAL3 [62]	18.05/0.527	21.51/0.759	23.03/0.862
BPFA-CS [64]	19.12/0.547	23.87/0.757	27.47/0.827
BCS-SPL [63]	20.29/0.542	24.53/0.733	27.46/0.841
BCS-FICTA [17] via the proposed ACT	**20.89/0.603**	**25.11/0.787**	**27.87/0.885**

among the competing methods, we have carefully tuned their parameters to achieve the best performance. Also, for the sake of fair comparison, the same test conditions are used in all experiments; i.e., the same sensing matrices are applied for BCS-SPL, BCS-FICTA, TVNLR, MH, RCoS, CS-ALS and the proposed CS-JASR.

We commence with demonstrating the effectiveness of utilizing the proposed ACT in promoting the efficiency of BCS-FICTA [17]. To this end, we tuned the parameters as follows: $k_{max} = 60$ and $\mathrm{TOL} = 10^{-3}$ (see Table II). Table IV lists the average PSNR and SSIM values of the recovered test images obtained by BCS-FICTA via the proposed ACT compared to those obtained by TVAL3, BPFA-CS and BCS-SPL (best results are emphasized in bold face). From Table IV, it can be inferred that the performance of BCS-FICTA via the proposed ACT is superior to those of compared algorithms in terms of both employed objective quality assessors. In the following, the effectiveness of CS-JASR is investigated. In our implementation, in process of finding NSSM for self-similarity in 3D transform domain, the size of each patch (i.e., $\sqrt{B_p} \times \sqrt{B_p}$) is set to 8×8. The size of training window for searching matched blocks (i.e., $\omega \times \omega$) is set to 20×20, and the number of best matched blocks (i.e., c) is set to 10. The other main parameters of the proposed algorithm are empirically set: $k'_{max} = 35$, $\mu_1 = \zeta\mu$, $\mu_2 = (1 - \zeta)\mu$, where the value of ζ typically varies in a range $[0.05, 0.3]$ and $\mu = 2.7 \times 10^{-3}$. Also, the orthogonal 3D transform denoted by $\mathcal{T}^{3D}$ is composed of 2D Bior 1.5 and 1D Haar transform. Table V lists the average PSNR and SSIM results of the recovered images by competing CS image recovery methods, for 10%, 30% and 50% of measurements. From Table V, we can see that TVNLR obtains the lowest PSNR and SSIM among all the compared algorithms. Moreover, the proposed CS-JASR considerably outperforms the other competing methods in most cases. The average PSNR (SSIM) gain of CS-JASR over the second best method, i.e., MARX-PC, can be up to 1.17 dB (0.137), at $\mathrm{MR} = 0.1$. Also, the average PSNR gain of CS-JASR over TVNLR, NLCS, MH and RCoS can be up to 4.37 dB, 3.61 dB, 2.51 dB and 1.62 dB, respectively.

For visual comparison, some CS recovered images by the competing methods are shown in Figs. 7 and 8. Better visual comparison can be made by zooming the images on the screen. Although the CS recovered images by MARX-PC, RCoS and MH achieve much better visual quality than those of NLCS, TVNLR and the other aforementioned competitive techniques, each of which still suffer from some undesirable artifacts (such as ringing and jaggy artifacts) and lost details. However, it can

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

TABLE V: PSNR/SSIM COMPARISONS OF THE PROPOSED CS-JASR WITH VARIOUS CS IMAGE RECOVERY METHODS.

		Algorithm							
Test Image	MR	TVNLR [42]	NLCS [46]	MH [65]	RCoS [48]	MARX-PC [45]	CS-ALS [67]	WSR-μ [68]	CS-JASR
Barbara	0.05	21.60/0.535	21.46/0.493	18.34/0.428	15.98/0.404	18.03/0.423	20.68/0.538	**22.57**/0.592	21.94/**0.621**
	0.1	22.48/0.622	22.72/0.562	27.46/0.833	23.77/0.683	24.25/0.652	26.75/0.828	27.06/0.796	**27.75/0.849**
	0.15	23.20/0.683	23.78/0.625	29.52/0.871	25.61/0.824	25.14/0.706	28.87/0.862	**30.33**/0.886	29.66/**0.892**
Baboon	0.05	18.12/0.286	18.46/0.263	14.72/0.180	14.55/0.214	14.80/0.185	16.06/0.260	17.93/0.262	**18.56/0.336**
	0.1	18.92/0.359	19.13/0.326	18.45/0.360	18.41/0.347	18.54/0.365	19.10/0.369	19.09/0.356	**19.47/0.400**
	0.15	19.18/0.416	19.69/0.385	19.46/0.426	19.33/0.400	19.57/0.432	19.96/0.449	19.84/0.439	**20.23/0.477**
Boat	0.05	22.36/0.611	22.52/0.595	20.23/0.504	24.43/0.676	22.73/0.629	23.11/0.641	24.81/0.675	**25.03/0.733**
	0.1	25.49/0.743	25.25/0.710	26.39/0.759	27.39/0.801	27.80/0.788	28.28/0.829	28.50/0.813	**28.59/0.846**
	0.15	27.70/0.810	27.18/0.775	28.60/0.828	29.48/0.864	29.95/0.845	30.21/0.868	31.09/0.884	**31.14/0.900**
House	0.05	24.50/0.737	26.64/0.763	25.28/0.721	26.29/0.752	28.55/0.784	27.55/0.790	29.87/0.819	**29.94/0.838**
	0.1	29.74/0.834	30.27/0.822	29.60/0.815	30.75/0.847	32.90/0.845	32.16/0.860	33.18/0.860	**33.20/0.875**
	0.15	31.84/0.864	32.24/0.853	32.64/0.874	34.11/0.888	34.15/0.867	34.02/0.881	**35.12**/0.887	34.99/**0.908**
Fence	0.05	18.41/0.439	18.62/0.388	14.80/0.329	14.86/0.315	15.64/0.332	16.10/0.442	17.08/0.469	**17.28/0.510**
	0.1	19.58/0.560	20.18/0.494	24.09/0.749	24.13/0.704	25.36/0.755	25.07/0.766	25.57/0.758	**25.69/0.791**
	0.15	21.17/0.649	21.83/0.590	26.42/0.812	26.50/0.803	26.94/0.818	27.33/0.829	28.23/0.838	**28.28/0.852**
Monarch	0.05	18.35/0.581	19.51/0.588	18.02/0.546	19.71/0.616	**21.89**/0.688	19.98/0.643	21.11/0.674	21.38/**0.733**
	0.1	22.99/0.773	22.97/0.724	23.05/0.762	24.20/0.802	**27.11**/0.839	25.26/0.837	26.48/0.837	26.01/**0.866**
	0.15	25.45/0.842	25.76/0.812	25.62/0.833	26.53/0.843	**29.14**/0.874	27.92/0.892	29.09/0.896	28.81/**0.911**
Starfish	0.05	21.02/0.545	21.35/0.535	18.97/0.460	21.64/0.555	21.19/0.535	21.35/0.569	21.45/0.572	**22.10/0.615**
	0.1	23.69/0.672	23.58/0.639	22.26/0.650	24.52/0.685	24.35/0.641	24.17/0.691	25.07/0.721	**25.40/0.737**
	0.15	25.17/0.744	25.11/0.710	25.41/0.742	26.91/0.764	25.49/0.724	26.79/0.789	27.54/0.807	**27.72/0.818**
Leaves	0.05	16.12/0.488	15.84/0.442	15.11/0.469	16.88/0.614	16.12/0.558	16.63/0.592	15.33/0.544	**17.85/0.668**
	0.1	19.42/0.712	18.46/0.611	20.44/0.720	25.48/0.873	24.27/0.793	25.13/0.851	22.75/0.804	**25.84/0.917**
	0.15	22.10/0.825	20.31/0.708	23.51/0.822	25.75/0.921	26.99/0.857	26.43/0.920	26.93/0.905	**27.17/0.939**
Straw	0.05	18.25/0.248	18.14/0.243	13.74/0.207	14.33/0.232	13.85/0.202	15.11/0.257	17.65/0.281	**19.22/0.385**
	0.1	18.85/0.362	19.08/0.359	20.37/0.621	19.69/0.431	19.00/0.322	20.76/0.574	19.88/0.511	**21.27/0.633**
	0.15	19.52/0.452	19.84/0.454	21.51/0.730	20.86/0.471	20.68/0.429	22.15/0.712	21.81/0.670	**22.78/0.742**
Texture	0.05	9.78/0.156	9.07/0.184	9.54/0.281	9.21/0.229	8.89/0.140	9.89/0.301	9.66/0.205	**10.57/0.352**
	0.1	11.94/0.328	11.91/0.332	13.01/0.620	12.02/0.414	11.31/0.212	13.10/0.628	11.52/0.355	**13.36/0.664**
	0.15	12.98/0.490	12.74/0.457	15.03/0.739	13.25/0.588	12.33/0.381	15.08/0.743	12.67/0.497	**15.62/0.766**
Average	0.05	18.85/0.463	19.16/0.449	16.87/0.412	17.79/0.461	18.17/0.448	18.65/0.503	19.75/0.509	**20.39/0.579**
	0.1	21.31/0.596	21.35/0.558	22.51/0.700	23.04/0.660	23.49/0.621	23.98/0.724	23.91/0.681	**24.66/0.758**
	0.15	22.83/0.677	22.85/0.637	24.77/0.768	24.83/0.737	25.04/0.693	25.88/0.794	26.26/0.771	**26.64/0.821**
	0.2	24.03/0.736	24.17/0.700	26.22/0.812	26.44/0.803	27.08/0.802	27.94/0.830	28.12/0.827	**28.33/0.853**
	0.3	26.11/0.811	26.57/0.823	28.29/0.858	28.80/0.859	29.32/0.850	30.12/0.890	30.16/0.889	**30.18/0.894**
	0.4	28.07/0.865	28.80/0.854	29.74/0.894	30.21/0.902	31.65/0.909	33.20/0.923	**33.25**/0.924	33.22/**0.927**
	0.5	29.70/0.899	31.08/0.879	31.56/0.919	32.67/0.927	33.20/0.929	**34.18**/0.945	34.12/0.944	34.07/**0.947**

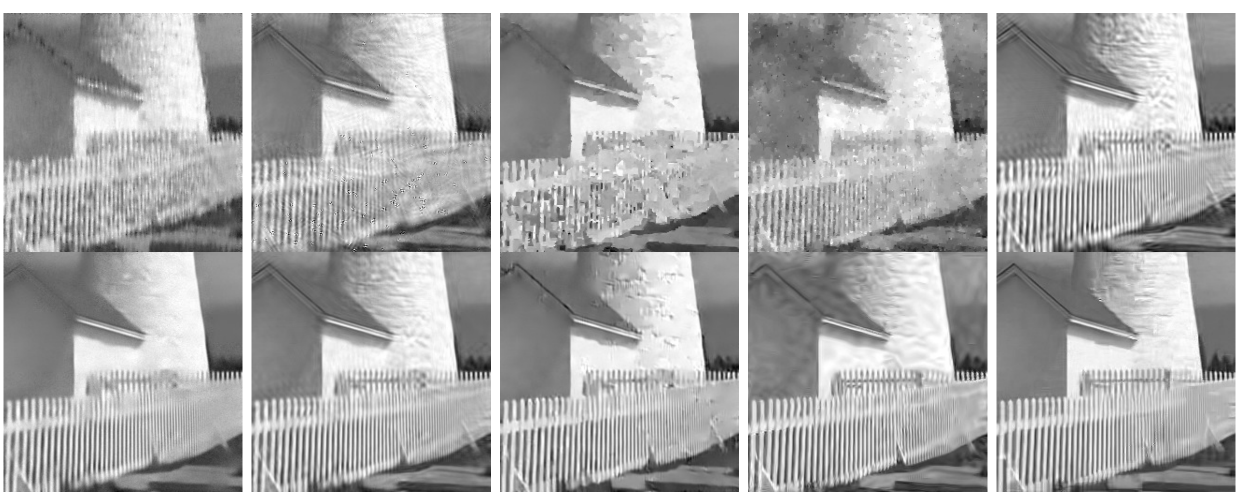


Fig. 7. CS recovered *Fence* images with 10% of measurements. Recovered results by: (a) BPFA-CS [64] (PSNR=19.95 dB, SSIM=0.566); (b) BCS-FICTA [17] via the proposed ACT (PSNR=20.02 dB, SSIM=0.585); (c) TVNLR [42]; (d) NLCS [46]; (e) MH [65]; (f) RCoS [48]; (g) MARX-PC [45]; (h) CS-ALS [67]; (i) WSR-μ [68]; and (j) the proposed CS-JASR. For quantitative comparison, see Table V.

be observed that the proposed CS-JASR produces not only sharp large-scale edges but also fine-scale image details. It preserves the image structures and textures better than the other competing methods and reproducing clearer images. The high performance of the proposed algorithm is attributed to the efficiently employment of both locally adaptive sparsity and nonlocal self-similarity constraint in the proposed JASR model offering a powerful mechanism for characterizing the

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

Fig. 8. CS recovered *Boat* images with 10% of measurements. Recovered results by: (a) BPFA-CS [64] (PSNR=24.67 dB, SSIM=0.705); (b) BCS-FICTA [17] via the proposed ACT (PSNR=24.90 dB, SSIM=0.720); (c) TVNLR [42]; (d) NLCS [46]; (e) MH [65]; (f) RCoS [48]; (g) MARX-PC [45]; (h) CS-ALS [67]; (i) WSR-μ [68]; and (j) the proposed CS-JASR. For quantitative comparison, see Table V.

TABLE VI: PSNR/SSIM COMPARISONS OF THE RECONSTRUCTED IMAGES FROM NOISY CS MEASUREMENTS UNDER DIFFERENT LEVELS OF GAUSSIAN NOISE, WITH MR=0.2.

		Algorithm		
Image	σ_e	CS-ALS [67]	WSR-μ [68]	CS-JASR
House	10	29.69/0.744	28.75/0.678	**30.87/0.769**
	20	24.77/0.535	24.03/0.444	**26.03/0.568**
	30	21.55/0.391	20.78/0.304	**22.95/0.410**
Monarch	10	26.11/0.787	26.38/0.688	**26.78/0.818**
	20	22.27/0.595	22.54/0.585	**23.41/0.644**
	30	19.67/0.473	19.66/0.450	**20.94/0.505**
Leaves	10	24.90/0.851	25.21/0.867	**25.71/0.897**
	20	21.09/0.701	21.42/0.721	**22.36/0.754**
	30	18.52/0.583	18.64/0.597	**19.86/0.633**

TABLE VII: PSNR/SSIM & NSSM TIME (seconds) OF THE RECOVERED *Barbara* AND *Monarch* IMAGES BY THE PROPOSED METHOD WITH TWO DIFFERENT 3D TRANSFORMS. THE NSSM TIME CORRESPONDS TO THE TIME CONSUMED FOR SOLVING THE $\boldsymbol{w}$ *sub-problem* OF (20c) IN EACH ITERATION. THOSE OF NSSM TIME ARE REPORTED IN PARENTHESES.

	Barbara		Monarch	
$\mathcal{T}^{3D}$	MR=0.2	MR=0.4	MR=0.2	MR=0.4
2D DCT +1D Haar	30.95/0.913 (8.333)	36.31/0.970 (8.855)	29.53/0.925 (8.423)	34.71/0.967 (8.978)
2D Bior 1.5 +1D Haar	**31.21/0.917** (**7.796**)	**36.45/0.972** (**8.212**)	**29.88/0.931** (**8.116**)	**34.98/0.971** (**8.744**)

structured sparsities of natural images.

B. Algorithm Robustness

In this section, we conducted some experiments with noisy CS measurements to demonstrate the robustness of the proposed CS-JASR to the noise. Toward this approach, a significant amount of AWGN, i.e., $\boldsymbol{e}$, is added to the CS measurements, see (1). In this case, the measurement ratio is fixed with $\mathrm{MR} = 0.2$, and the standard deviation of the Gaussian noise, i.e., $\sigma_{\boldsymbol{e}}$ is varied from 10 to 30. Also, in this experiment, we just consider the two superior state-of-the-art methods of CS-ALS and WSR-μ for comparison. The PSNR and SSIM comparison of the reconstructed *House*, *Monarch*, and *Leaves* images are shown in Table VI. As can be seen, the proposed CS-JASR outperforms both CS-ALS and WSR-μ in each situation. This experiment was run on the other test images with different measurement ratios under different levels of Gaussian noise, and the obtained results confirmed the robustness and efficiency of the proposed method in the presence of the noise.

C. Discussions: Comparison Between ℓ_0 and ℓ_1 Minimization, Algorithm Convergence, and The 3D Transform $\mathcal{T}^{3D}$

As stated previously, in the proposed JASR prior described in (17), the nonlocal self-similarity constraint is obtained by summing all the ℓ_1 norm of the transform coefficients (see (3)). Hence, the related sub-problem described in (25) is solved via soft-thresholding. However, if the nonlocal self-similarity constraint was written as sum of all ℓ_0 pseudo-norm of the transform coefficients (number of nonzero coefficients), it would be solved via hard-thresholding—similar to that performed in [48]. In this experiment, we make a comparison between using ℓ_1 and ℓ_0 minimization in solving (25) for the proposed method. For comparing the efficiency of utilizing ℓ_1 norm over ℓ_0 pseudo-norm in achieving the nonlocal self-similarity constraint, Fig. 9 plots some progression curves of PSNR versus the iteration number, just by considering $k'_{max} = 80$ as the stopping criterion. As can be seen, the results obtained by using ℓ_1 norm (black solid line curve; the proposed CS-JASR) has better performance than those obtained by using ℓ_0 pseudo-norm (red dashed line curve). This corroborates the convenience and superiority of using ℓ_1 over ℓ_0 in the used nonlocal self-similarity constraint, validating the effectiveness of our approach to solve (25) via soft-thresholding.

Furthermore, for the convergence study of the proposed method, we only provide empirical evidence to illustrate it. From Fig. 10, it can be observed that with the growth of iteration number, all the PSNR curves increase monotonically and ultimately become flat and stable, which fully demonstrates the convergence of the proposed algorithm. This fact motivated

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TIP.2016.2562563, IEEE Transactions on Image Processing

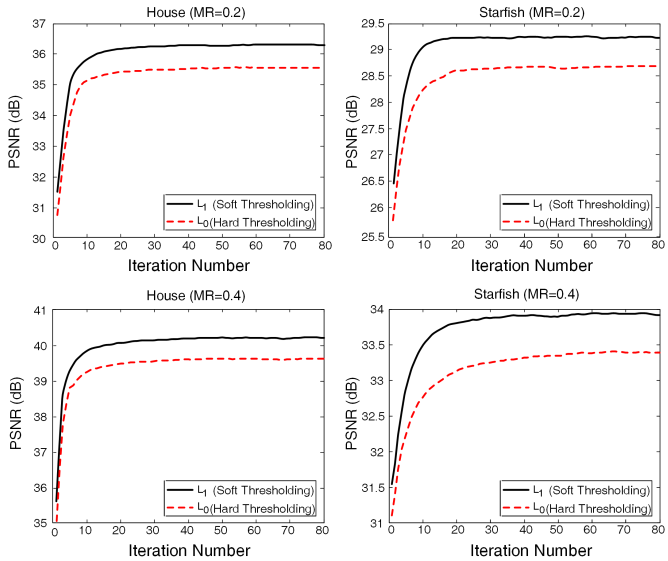


Fig. 9. Comparison between the efficiency of utilizing ℓ_1 and ℓ_0 for finding the nonlocal self-similarity constraint, in the proposed JASR prior. Progression of the PSNR vs. iteration number, for images *House* $\mathrm{MR} \in \{0.2, 0.4\}$ (left plots), and *Starfish* $\mathrm{MR} \in \{0.2, 0.4\}$ (right plots).

us to introduce the second stopping criterion, as illustrated in Table III, to prevent further iterations after convergence of the algorithm. Our experiments show that the proposed CS-JASR usually converges in around 20 iterations.

As mentioned previously, the 3D transform used in JSM [20] and CoSM [48] is composed of 2D DCT and 1D Haar transform; however, in this paper, we employ a $\mathcal{T}^{3D}$ which is composed of 2D Bior 1.5 and 1D Haar transform. The results corresponding to effects of using these two various $\mathcal{T}^{3D}$ are presented in Table VII. As can be seen, the results obtained by utilizing the later $\mathcal{T}^{3D}$ are slightly better than the first one, which fully demonstrates the convenience of this adoption.

D. Computational Complexity and Running Time

The computational cost of the proposed CS-JASR algorithm (illustrated in Table III) mainly comes from three sources:

1) $\boldsymbol{\vartheta}$ *sub-problem*: estimating the curvelet coefficients of original image using available perturbed coefficients: ACT—step (1) to step (3).
2) $\boldsymbol{w}$ *sub-problem*: finding similar patches by searching within search window, stacking into a 3D array, the operation of 3D transform, soft thresholding, the operation of inverse 3D transform, and the final averaging—step (4) to step (6).
3) $\boldsymbol{u}$ *sub-problem*: employing the steepest descent method for solving the minimization of the quadratic function in (20a)—step (7).

In evaluating the computational complexity of the $\boldsymbol{\vartheta}$ *sub-problem*, ACT utilizes 3 steps of executing the forward curvelet transform, performing the soft-thresholding on each element (curvelet coefficient), and finally, obtaining $\boldsymbol{\vartheta}$ by concatenating all elements. The overall cost of all these steps along with obtaining the inverse curvelet transform of estimated $\boldsymbol{\vartheta}$ is $\mathcal{O}(n' + n \log n)$, where n' is the total number of elements used in shrinkage step, and n is the size of original image (i.e., $\sqrt{n} \times \sqrt{n}$). Based on the experiments, for an image of size 256×256, $\boldsymbol{\vartheta}$ *sub-problem* consumes 0.4 second on average, in each iteration. Solving $\boldsymbol{w}$ *sub-problem* needs $\mathcal{O}\big(PB_p(c + c\log B_p + \omega^2 + \log \omega^2)\big)$ operations, where $P = (\sqrt{n} - \sqrt{B_p} + 1)^2$ is the number of all overlapped patches $\boldsymbol{u}_{p_{l'}}$ of size $\sqrt{B_p} \times \sqrt{B_p}$ extracted from the available approximated image $\boldsymbol{z}$, and c is the number of best matched patches in a search window of size $\omega \times \omega$. Also, the elapsed time for solving $\boldsymbol{w}$ *sub-problem* is about 8.3 seconds, in each iteration. The steepest descent step needs $\mathcal{O}\Big(J_0\big(n^3 + n(m+4+\log n)\big)\Big)$ operations, where J_0 denotes the number of inloop iterations, and m stands for the length measurement vector $\boldsymbol{f}$. The elapsed time for computing the steepest descent in each inloop iteration is about 0.02 second.

TABLE VIII: Running Time Comparison Between the Proposed CS-JASR and the Other Competing State-of-the-Art Methods on Image *House* of Size 256×256 With $\mathrm{MR} = 0.2$ (unit: second).

Algorithm	PSNR (dB)	SSIM	Time (sec)
TVNLR [42]	33.12	0.879	147
NLCS [46]	33.52	0.873	14
MH [65]	33.79	0.882	21
RCoS [48]	35.15	0.894	1170
MARX-PC [45]	35.37	0.885	633
CS-ALS [67]	35.56	0.904	217
WSR-μ [68]	36.39	0.909	245
CS-JASR	36.28	0.916	179

A running time comparison between the proposed CS-JASR and the other competing state-of-the-art methods in CS image recovery is shown in Table VIII. Based on the experiments, under mentioned environment, BCS-FICTA via the proposed ACT and the proposed CS-JASR require about 6$\sim$16 seconds and 2$\sim$4 minutes to recover a CS image of size 256×256, respectively.

VII. Conclusions

In this paper, we have proposed an adaptive curvelet thresholding (ACT) criterion for adaptively eliminating perturbations which appear during CS recovery process. Also, we have extended our previous work [17] by employing ACT within its framework to promote its efficiencies, leading to a simple and effective CS image recovery method. Furthermore, we have introduced a novel sparsity measure called joint adaptive sparsity regularization (JASR), and a new strategy for high-fidelity CS image restoration via JASR. The proposed JASR efficiently characterizes the intrinsic sparsities of natural images by exploiting both local sparsity and nonlocal 3D sparsity, simultaneously. Also, we demonstrated the superiority of utilizing ℓ_1 norm over ℓ_0 pseudo-norm in obtaining the nonlocal statistical self-similarity constraint, NSSM. In addition, we showed that, utilizing a $\mathcal{T}^{3D}$ composed of 2D Bior 1.5 and 1D Haar transform gains better performance than utilizing a $\mathcal{T}^{3D}$ composed of 2D DCT and 1D Haar transform. Our comprehensive experimental results demonstrated that the proposed methods significantly outperform the conventional and the current state-of-the-art CS image recovery methods in terms of both quantitative metrics and subjective visual quality. Moreover, our proposed methods exhibit very good convergence property.

For future work, due to the *ill-posed* nature of image restoration, we would like to extend the proposed sparsity measure, JASR, to the other image restoration applications, e.g., denoising, inpainting, deblurring, and super-resolution.

VIII. ACKNOWLEDGEMENT

The authors would like thank the associate editor for his efforts in coordinating the review of the manuscript, and also would like to thank the three anonymous reviewers for their constructive comments and helpful suggestions to improve the manuscript. They also would like to express their gratitude to Prof. Jalal Fadili (CNRS-ENSICAEN-Universitè de Caen) and Dr. Jian Zhang (Peking University) for many fruitful discussions; and the authors of [20], [42], [45], [46], [48], [62], [63], [64], [65], [67], and [68] for sharing the source code of their papers used in Section VI.

REFERENCES

[1] E. Candès, J. Romberg, and T. Tao, "Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information," *IEEE Trans. Inf. Theory*, vol. 52, no. 2, pp. 489-509, Feb. 2006.

[2] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289-1306, Apr. 2006.

[3] L. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D.*, vol. 60, pp. 259-268, Nov. 1992.

[4] L. Bregman, "The relaxation method of finding the common points of convex sets and its application to the solution of problems in convex programming," *Comput. Math. Math. Phys.*, vol. 7, pp. 200-217, 1967.

[5] I. Daubechies, M. Defrise, and C. D. Mol, "An iterative thresholding algorithm for linear inverse problems with a sparsity constraint," *Commun. Pure. Appl. Math.*, vol. 57, no. 11, pp. 1413-1457, Nov. 2004.

[6] A. Beck and M. Tebulle, "A fast iterative shrinkage-thresholding algorithm for linear inverse problems," *SIAM J. Imaging Sci.*, vol. 2, no. 1, pp. 183-202, Jan. 2009.

[7] J. Bioucas-Dias, and M. Figueiredo, "A new TwIST: two step iterative shrinkage/thresholding algorithms for image restoration," *IEEE Trans. Image Process.*, vol. 16, no. 12, pp. 2992-3004, Dec. 2007.

[8] T. Goldstein and S. Osher, "The split Bregman method for ℓ_1 regularized problems," *SIAM J. Imag. Sci.*, vol. 2, no. 2, pp. 323-343, Apr. 2009.

[9] W. Yin, S. Osher, D. Goldfarm, and J. Darbon, "Bregman iterative algorithms for ℓ_1 minimization with applications to compressed sensing," *SIAM J. Imag. Sci.*, vol. 1, no. 1, pp. 143-168, Mar. 2008.

[10] M. Elad, P. Milanfar, and R. Rubinstein, "Analysis versus synthesis in signal priors," *Inverse Problems*, vol. 23, no. 3, pp. 947968, Apr. 2007.

[11] E. J. Candès and D. L. Donoho, "Ridgelets: A key to higher-dimensional intermittency?" *Phil. Trans. R. Soc. London A*, vol. 357, no. 1760, pp. 2495-2509, Sep. 1999.

[12] E. Candès and D. Donoho, "Curvelets—A surprisingly effective non-adaptive representation for objects with edges, in *Curves and Surface Fitting: Saint-Malo 1999*, A. Cohen, C. Rabut, and L. Schumaker, Eds. Nashville: Vanderbilt Univ. Press, 2000, pp. 105-120.

[13] M. N. Do and M. Vetterli, "The contourlet transform: An efficient directional multiresolution image representation," *IEEE Trans. Image Process.*, vol. 14, no. 12, pp. 2091-2106, Dec. 2005.

[14] E. Candès and D. Donoho, "New tight frames of curvelets and optimal representations of objects with piecewise singularities," *Commun. Pure Appl. Math.*, vol. 57, no. 2, pp. 219266, Feb. 2004.

[15] J. Ma and G. Plonka,"The curvelet transform," *IEEE Signal Process. Mag.*, vol. 72, no. 2, pp. 118-133, Mar. 2010.

[16] J. Ma, "Improved iterative curvelet thresholding for compressed sensing," *IEEE Trans. Inst. Measure.*, vol. 60, no. 1, pp. 126-136, Jan. 2011.

[17] N. Eslahi, A. Aghagolzadeh, and S. M. H. Andargoli, "Block compressed sensing images using curvelet transform," in *Proc. 22nd Iranian Conf. Electr. Eng. (ICEE)*, Tehran, Iran, May 2014, pp. 1581-1586.

[18] A. N. Tikhonov and V. Y. Arsenin, *Solutions of ill-posed problems*, Washington, D.C., USA: V. H. Winston & Sons, 1977.

[19] X. Li "Image recovery via hybrid sparse representation: a deterministic annealing approach," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 5, pp. 953-962, Sep. 2011.

[20] J. Zhang, D. Zhao, R. Xiong, S. Ma, and W. Gao, "Image restoration using joint statistical modeling in space-transform domain," *IEEE Trans. Circuits Syst. video Technol.*, vol. 24, no. 6, pp. 915-928, Jun. 2014.

[21] S. Lefkimmiatis and S. Osher, "Non-local structure tensor functionals for image regularization," *IEEE Trans. Comput. Imag.*, vol. 1, no. 1, pp. 16-29, Mar. 2015.

[22] A. Buades, B. Coll, and J. M. Morel, "A review of image denoising algorithms, with a new one," *SIAM Multiscale Model. Sim.*, vol. 4, no. 2, pp. 490-530, 2005.

[23] G. Peyrè, S. Bougleux, and L. D. Cohen, "Non-local regularization of inverse problems," *Inverse Prob. Imag.*, vol. 5, no. 2, pp. 511-530, 2011.

[24] M. Jung, X. Bresson, T. F. Chan, and L. A. Vese, "Nonlocal Mumford-Shah regularizers for color image restoration," *IEEE Trans. Image Process.*, vol. 20, no. 6, pp. 1583-1598, Jun. 2011.

[25] W. Dong, L. Zhang, G. Shi, and X. Wu, "Image deblurring and super-resolution by adaptive sparse domain selection and adaptive regularization," *IEEE Trans. Image Process.*, vol. 20, no. 7, pp. 1838-1857, Jul. 2011.

[26] K. Zhang, X. Gao, D. Tao, and X. Li, "Single image super-resolution with non-local means and steering kernel regression," *IEEE Trans. Image Process.*, vol. 21, no. 11, pp. 4544-4556, Nov. 2012.

[27] W. Dong, L. Zhang, R. Lukac, and G. Shi, "Sparse representation based image interpolation with nonlocal autoregressive modeling," *IEEE Trans. Image Process.*, vol. 22, no. 4, pp. 1382-1394, Apr. 2013.

[28] W. Dong, L. Zhang, G. Shi, and X. Li, "Nonlocally centralized sparse representation for image restoration," *IEEE Trans. Image Process.*, vol. 22, no. 4, pp. 1620-1630, Apr. 2013.

[29] N. Wang, D. Tao, X. Gao, X. Li, and J. Li, "Transductive face sketch-photo synthesis," *IEEE Trans. Neural Netw. Learn. Syst.,* vol. 24, no. 9, Sep. 2013.

[30] J. Yu, Y. Rui, and D. Tao, "Click prediction for web image reranking using multimodal sparse coding," *IEEE Trans. Image Process.*, vol. 23, no. 5, pp. 2019-2032, May 2014.

[31] J. Jiang, L. Zhang, and J. Yang, "Mixed noise removal by weighted encoding with sparse nonlocal regularization," *IEEE Trans. Image Process.*, vol. 23, no. 6, pp. 2651-2662, Jun. 2014.

[32] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862-2869.

[33] U. Schmidt and S. Roth, "Shrinkage fields for effective image restoration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2774-2781.

[34] Y. Romano, M. Protter, and M. Elad, "Single image interpolation via adaptive nonlocal sparsity-based modeling," *IEEE Trans. Image Process.*, vol. 23, no. 7, pp. 3085-3098, Jul. 2014.

[35] W. Dong, G. Shi and X. Hu, and Y. Ma, "Nonlocal sparse and low-rank regularization for optical flow estimation," *IEEE Trans. Image Process.*, vol. 23, no. 10, pp. 4527-4538, Oct. 2014.

[36] V. Estellers, S. Soatto, and X. Bresson, "Adaptive regularization with the structure tensor," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1777-1790, Jun. 2015.

[37] X. Li, H. He, R. Wang, and D. Tao, "Single image superresolution via directional group sparsity and directional features," *IEEE Trans. Image Process.*, vol. 24, no. 9, pp. 2874-2888, Sep. 2015.

[38] C. Deng, J. Xu, K. Zhang, D. Tao, X. Gao, and X. Li, "Similarity constraints-based structured output regression machine: an approach to Image super-resolution," *To appear in IEEE Trans. Neural Netw. Learn. Syst.,* 2015, DOI: 10.1109/TNNLS.2015.2468069.

[39] R. Wang and D. Tao, "Non-local auto-encoder with collaborative stabilization for image restoration," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2117-2129, May 2016.

[40] X. Zhang, M. Burger, X. Bresson, and S. Osher, "Bregmanized nonlocal regularization for deconvolution and sparse reconstruction," *SIAM J. Imaging Sci.*, vol. 3, no. 3, pp. 253-276, 2010.

[41] W. Dong, X. Yang, and G. Shi, "Compressive sensing via reweighted TV and nonlocal sparsity regularisation," *IET Electronics lett.*, vol. 49, no. 3, pp. 184-186, Jan. 2013.

[42] J. Zhang, S. Liu, R. Xiong, S. Ma, and D. Zhao, "Improved total variation based image compressive sensing recovery by nonlocal regularization," in *Proc. IEEE Int. Symp. Circuits and Systems*, May 2013, pp. 2836-2839.

[43] W. Dong, G. Shi, X. Wu, and L. Zhang, "A learning-based method for compressive image recovery," *J. Vis. Commun. Image R.*, vol. 24, no. 7, pp. 1055-1063, Oct. 2013.

[44] W. Dong, G. Shi, X. Li, L. Zhang, and X. Wu, "Image reconstruction with locally adaptive sparsity and nonlocal robust regularization," *Signal Process. Image Commun.*, vol. 27, no. 10, pp. 1109-1122, Nov. 2012.

[45] X. Wu, W. Dong, X. Zhang, and G. Shi, "Model-assisted adaptive recovery of compressed sensing with imaging applications," *IEEE Trans. Image Process.*, vol. 21, no. 2, pp. 451-458, Feb. 2012.

[46] X. Shu, J. Yang, and N. Ahuja, "Non-local compressive sampling recovery," in *Proc. IEEE Int. Conf. Comput. Photography (ICCP)*, May 2014, pp. 1-8.

[47] W. Dong, G. Shi, X. Li, Y. Ma, and F. Huang, "Compressive sensing via nonlocal low-rank regularization," *IEEE Trans. Image Process.*, vol. 23, no. 8, pp. 3618-3632, Aug. 2014.

[48] J. Zhang, D. Zhao, C. Zhao, R. Xiong, S. Ma and W. Gao, "Image compressive sensing recovery via collaborative sparsity," *IEEE J. Emerg. Sel. Topics Circuits Syst.*, vol. 2, no. 3, pp. 380-391, Sep. 2012.

[49] S. Li and H. Qi, "A Douglas-Rachford splitting approach to compressed sensing image recovery using low-rank regularization," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4240-4249, Nov. 2015.

[50] E. Candès, L. Demanet, D. Donoho, and L. Ying, "Fast discrete curvelet transforms," *Multiscale Model. Simul.*, vol. 5, no. 3, pp. 861-899, Jan. 2006.

[51] Z. Wong, A. C. Bovik, H. R. Sheikh and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600-612, Apr. 2004.

[52] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080-2095, Aug. 2007.

[53] H. J. Trussell, and M. R. Civanlar, "The Landweber iteration and projection onto convex sets," *IEEE Trans. Acoust. Speech, Signal Process.*, vol. 33, no. 6, pp. 1632-1634, Dec. 1985.

[54] J. L. Starck, E. J. Candès, and D. L. Donoho, "The curvelet transform for image denoising," *IEEE Trans. Image. Process.*, vol. 11, no. 6, pp. 670-684 Jun. 2002.

[55] H. Rabbani and S. Gazor, "Image denoising employing local mixture models in sparse domains," *IET Image Process.*, vol. 4, no. 5, pp. 413-428, Oct. 2010.

[56] B. Zhang, J. Fadili, and J. Starck, "Wavelets, ridgelets, and curvelets for Poisson noise removal," *IEEE Trans. Image Process.*, vol. 17, no. 7, pp. 1093-1108, Jul. 2008.

[57] S. Palakkal and K. M. M. Prabhu, "Poisson mage denoising using fast discrete curvelet transform and wave atom," *Signal Process.*, vol. 92, no. 9, pp. 2002-2017, Sep. 2012.

[58] L. Boubchir and M. J. Fadili, "Multivariate statistical modeling of images with the curvelet transform," in *proc. the 8th IEEE Int. Symp. Signal Process. and its Applications (ISSPA)*, Sydney, Australia, Aug. 2005, pp. 747-750.

[59] D. L. Donoho and I. M. Johnstone, "Ideal spatial adaptation by wavelet shrinkage," *Biometrika*, vol. 81, pp. 425-455, Aug. 1994.

[60] M. K. Mihçak, I. Kozintsev, K. Ramchandran, and P. Moulin, "Low-complexity image denoising based on statistical modeling of wavelet coefficients," *IEEE Signal Process. Lett.*, vol. 6, no. 12, Dec. 1999.

[61] D. L. Donoho, "De-noising by soft-thresholding," *IEEE Trans. Inf. Theory.*, vol. 41, no. 3, pp. 613-627, May. 1995.

[62] C. Li, W. Win, H. Jing, and Y. Zhang, "An efficient augmented Lagrangian method with applications to total variation minimization," *Comput. Optim. Appl.*, vol. 56, no. 3, pp. 507-530, Dec. 2013.

[63] S. Mun, and J. E. Fowler, "Block compressed sensing of images using directional transforms," in *Proc. 16th IEEE Int. Conf. Image Process. (ICIP)*, Nov. 2009, pp. 3021-3024.

[64] M. Zhou, H. Chen, J. Paisely, L. Ren, L. Li, Z. Xing, D. Dunson, G. Sapiro, and L. Carin, "Nonparametric Bayesian dictionary learning for analysis of noisy and incomplete images," *IEEE Trans. Image Process.*, vol. 21, no. 1, Jan. 2012.

[65] C. Chen, E. W. Tramel, and J. E. Fowler, "Compressed sensing recovery of images and video using multihypothesis predictions," in *Proc. 45th Asilomar Conf. Signals Sys. Comput.*, Nov. 2011, pp. 1193-1198.

[66] N. Eslahi, M. Mahdavinataj, A. Aghagolzadeh, "Mixed Gaussian-impulse noise removal from highly corrupted images via adaptive local and nonlocal statistical priors," in *Proc. 9th Iranian Mach. Vis. Image Process. (MVIP)*, Tehran, Iran, Nov. 2015, pp. 70-75.

[67] J. Zhang, C. Zhao, D. Zhao, and W. Gao, "Image compressive sensing recovery using adaptively learned sparsifying basis via ℓ_0 minimization," *Signal Process.*, vol. 103, pp. 114-126, Oct. 2014.

[68] W. Dong, X. Wu, and G. Shi, "Sparsity fine tuning in wavelet domain with application to compressive image reconstruction," *IEEE Trans. Image Process.*, vol. 23, no. 12, Dec. 2014.

Nasser Eslahi (S'14) received the B.S. and M.S. degrees in electrical engineering from Imam Khomeini International University and Babol University of Technology, Iran, in 2012 and 2015, respectively. He is currently working toward his Ph.D. degree in the Department of Signal Processing, Tampere University of Technology, Tampere, Finland.

His research interests include image and video processing, statistical signal processing, sparse representation/approximation, compressive sensing, image inverse problems and convex optimization.

Ali Aghagolzadeh (S'87-M'92-SM'08) received the B.S. degree in electrical and electronic engineering from Tabriz University, Tabriz, Iran, in 1985. He received the M.S. and the Ph.D. degrees in electrical engineering from the Illinois Institute of Technology, Chicago, IL, USA, and Purdue University, West Lafayette, IN, USA, in 1988 and 1991, respectively.

He is currently a Professor with the Department of Electrical and Computer Engineering, Babol University of Technology, Babol, Iran. His research interests include image processing, video coding and compression, information theory, and computer vision. He has supervised numerous master and Ph.D. students and published more than 200 scientific peer-reviewed journal and conference papers.